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MISO DPP 2021 Cycle East (ATC) Area Study Phase 3 Final Report

Prepared for MISO

Submitted By:

ATC Resource Interconnections

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1.0 EXECUTIVE SUMMARY

Sixteen (16) resource projects have requested to interconnect to the MISO transmission network in the East (ATC) Area and are included in the Definitive Planning Phase 2021 Cycle Phase 3 study (DPP 2021 Cycle Phase 3). All Generating Facilities have requested both Energy Resource Interconnection Service (ERIS) and Network Resource Interconnection Service (NRIS).

This report presents the study results of a System Impact Study (SIS) performed to evaluate the interconnection of the generator interconnection requests in the DPP 2021 Cycle Phase 3 study. The study was performed under the direction of Midcontinent Independent System Operator (MISO) by ATC and an ad hoc study group. The results for 2026 scenario are summarized below.

1.1 Project List

The DPP 2021 Cycle Phase 3 has sixteen (16) interconnection requests with a combined 3,026.09 MW of ERIS and 2,382.37 MW NRIS. The DPP 2021 Cycle Phase 3 generator interconnection requests are listed in Table 1.1-1. The one-line diagrams of the interconnection facilities are shown in Appendix C.

Table 1.1-1 – Generating Facilities in DPP 2021 Cycle Phase 3 East (ATC) Area

MISO Queue #	Service Type	Control Area	County, State	Point of Interconnection	Fuel Type	Charging Election (MW) ²	ERIS (MW)	NRIS (MW)	Inverter Model	Main Step-Up Transformer ¹
J1895	NRIS	ALTE	Kenosha, WI	Balsam 138 kV	Battery	200	200	200	68 x 3.51 MVA PE FREEMAQ FP3510M3 Total MVA = 238.68	One 138/34.5/13.8 kV 225/250/275 MVA %Z = 11
J1931	NRIS	ALTE	Iowa, WI	Hill Valley 345 kV	Wind	N/A	40	12	16 x 3.13 MVA GE 2.8-127 Total MVA = 50.13	One 345/34.6/13.8 kV 30/40/50 MVA %Z = 9
J1934	NRIS	WEC	Kenosha, WI	Lakeview 138 kV	Battery	150	150	150	40 x 4.2 MVA TMEIC Ninja EES BSU-L0 840GR Total MVA = 168	One 138/34.5/13.8 kV 110/150/185 MVA %Z = 9
J1935	NRIS	ALTE	Dane, WI	North Madison 69 kV	Solar	N/A	50	50	14 x 4.2 MVA TMEIC Ninja PVU-L0 840GR Total MVA = 58.8	One 69/34.5/13.8 kV 40/50/65 MVA %Z = 9
J1943	NRIS	MGE	Columbia, WI	J1410/J1411 Interconnection Substation 345 kV (North Madison - Columbia 345 kV)	Solar	N/A	200	200	54 x 4.2 MVA TMEIC Ninja PVU-L0 840GR Total MVA = 226.8	Two 345/34.5/13.8 kV 75/100/125 MVA %Z = 9
J1993	NRIS	ALTE	Columbia, WI	J1629 Interconnection Substation 345 kV (Columbia - South Fond du Lac 345 kV)	Solar	N/A	99	99	28 x 4.2 MVA SMA SC 4200-UP-US Total MVA = 117.6	One 345/34.5/13.8 kV 66/88/110 MVA %Z = 9
J2029	NRIS	UPPC	Marquette, MI	Perch Lake 138 kV	Solar	N/A	62.5	62.5	21 x 3.6 MVA Sungrow SG3600D Total MVA = 75.6	Two 138/34.5/13.8 kV 24/40/40 MVA %Z = 9.5
J2055	NRIS	ALTE	Sheboygan, WI	Edgewater 345 kV	Solar	N/A	100	100	27 x 4.01 MVA PE HESM Gen 3 FS4010M Total MVA = 108.27	One 138/34.5/13.8 kV 72/95/120 MVA %Z = 10
J2099	NRIS	WPS	Portage, WI	J2099/J2107/J2185 Interconnection Substation 345 kV (Rocky Run - Werner West 345 kV)	Solar	N/A	399.03	195.79	120 x 3.55 MVA PE HEM FS3430M 06 Total MVA = 426	Two 345/34.5/13.8 kV 130/172/216 MVA %Z = 7.992
J2107	NRIS	WPS	Portage, WI	J2099/J2107/J2185 Interconnection Substation 345 kV (Rocky Run - Werner West 345 kV)	Solar	N/A	399.03	195.79	120 x 3.55 MVA PE HEM FS3430M 06 Total MVA = 426	Two 345/34.5/13.8 kV 130/172/216 MVA %Z = 7.992
J2148	NRIS	WEC	Outagamie, WI	Werner West 138 kV	Solar	N/A	150	150	43 x 4.2 MVA PE Freesun FS4200M Total MVA = 180.6	Shared with J2074
J2185	NRIS	WPS	Portage, WI	J2099/J2107/J2185 Interconnection Substation 345 kV (Rocky Run - Werner West 345 kV)	Solar	N/A	399.03	195.79	120 x 3.55 MVA PE HEM FS3430M 06 Total MVA = 426	Two 345/34.5/13.8 kV 130/172/216 MVA %Z = 7.992
J2193	NRIS	ALTE	Waushara, WI	Sand Lake 138 kV	Solar/Battery	25	100	100	29 x 3.6 MVA (solar) & 16 x 3.6 MVA (battery) Sungrow SG3600D Total MVA = 104.4 (solar) & 57.6 (battery)	One 138/34.5/13.8 kV 135/180/225 MVA %Z = 7.8
J2211	NRIS	WPS	Brown, WI	Tower Drive 138 kV	Battery	200	200	200	68 x 3.51 MVA PE Freemaq FP3510M3 Total MVA = 238.68	One 138/34.5/13.8 kV 225/250/275 MVA %Z = 11
J2304	NRIS	ALTE	Columbia, WI	Rockdale - Columbia 345 kV	Battery	400	400	400	136 x 3.365 MVA SMA SCS 3950 Total MVA = 457.5	Two 345/34.5/13.8 kV 134/178/222 MVA %Z = 11.5
J2316	NRIS	MIUP	Dickinson, MI	Nordic 138 kV	Solar	N/A	77.5	71.5	24 x 4.004 MVA SMA SC4400 UP-US Total MVA = 96.096	One 138/34.5/13.8 kV 90/120/150 MVA %Z = 9.5

¹Three winding transformers add a node not assigned to a PSSE bus number which must be initialized for every model build. Past experience has shown that these nodes not assigned to a PSSE bus number have been a source for power flow model solution difficulties. Since the third winding, when it has no devices such as a shunt or load connected to it, has no impact on the steady-state modeling ATC has opted to not model them. The three-winding transformers will be modeled as a three-winding transformer for the short circuit study

²Charging election indicates whether the storage device(s) will be charged using energy from the Transmission System at any time and the maximum MW that will be withdrawn from the Transmission System at any time. 0 MW indicates the storage device(s) will NOT be charged using energy from the Transmission System. This election may be reduced to mitigate constraints in the shoulder batteries charging scenario.

1.2 Generating Facility Requirements

1.2.1 Voltage Schedule Requirement

ATC requires all generators in its territory to maintain a voltage schedule at the Point of Interconnection (POI). The standard voltage schedule is 1.02 per unit as measured at the POI. This schedule may be changed by the Transmission Operator for specific power plants or specific conditions.

1.2.2 Power Factor Range Requirement

FERC Order 827 and ATC Criteria require all newly interconnecting generators interconnecting to ATC-owned Facilities to provide a power factor range for synchronous and non-synchronous (e.g., wind turbines, solar) generation of 0.95 leading (when a Generating Facility is consuming reactive power from the transmission system) to 0.95 lagging (when a Generating Facility is supplying reactive power to the transmission system).

The Generating Facility must be capable of maintaining ATC's standard power factor range at all power output levels by providing continuous dynamic reactive power at the following locations (i.e. point of measurement):

- A. The POI for all synchronous generators
- B. The high-side of the generator substation for all non-synchronous generators

This power factor requirement applies at all power output levels unless the Generating Facility is physically disconnected from the ATC transmission system. Physically disconnected means an open

- generator step-up transformer high or low-side breaker(s) for a synchronous Generation Facility, or
- generator substation step-up transformer high-side breaker(s) for a nonsynchronous Generation Facility.

For synchronous generators, the interconnection studies will account for the net effect of all energy production devices and losses on the Customer's side of the POI.

- Dynamic reactive power provided by a synchronous Generating Facility may meet the dynamic reactive power requirement by utilizing a combination of the inherent dynamic reactive power capability of the synchronous generator, dynamic reactive power devices (e.g., Static VAR Compensators), and static reactive power devices (e.g., capacitors) to make up for losses.
- Dynamic leading reactive power provided by a synchronous Generating Facility cannot use inductive losses from generator step-up transformer(s) and generator tie line(s) to meet the leading power factor calculation at POI. A synchronous Generating Facility must be able to meet a 0.95 leading power factor, as measured at the generator terminal (i.e. the low side of the generator step-up transformer).

For non-synchronous generators, the interconnection studies will account for the net effect of all energy production devices and losses on the Customer's side of generator substation step-up transformer.

- Dynamic reactive power provided by a non-synchronous Generating Facility must meet the following requirement from FERC order 827 paragraph 35:
 - "Non-synchronous generators may meet the dynamic reactive power requirement by utilizing a combination of the inherent dynamic reactive power capability of the inverter, dynamic reactive power devices (e.g., Static VAR Compensators), and static reactive power devices (e.g., capacitors) to make up for losses."
- Dynamic leading reactive power provided by a non-synchronous Generating Facility cannot use inductive losses from pad-mount and station step-up transformers and collector system to meet the leading power factor calculation at the high-side of the generator substation. A non-synchronous Generating Facility must be able to meet a leading 0.95 power factor, as measured at the generator terminal (i.e. the low side of the pad-mount transformer).
- Dynamic lagging reactive power provided by a non-synchronous Generating Facility cannot use collector system charging to meet the lagging power factor calculation at the high-side of the generator substation. A non-synchronous Generating Facility must be able to meet a lagging 0.95 power factor, as measured at the generator terminal (i.e. the low side of the pad-mount transformer).
- When the Generating Facility is not generating active power (i.e. zero MW output):
 - a. The reactive power injection to the transmission system at the high-side of the generator substation should be zero Mvar. Exceptions for a short transition period during the switching of inverter control modes may be granted on a case by case basis if there is no reliability concern. For example, solar PV facilities may have a transition period during sunrise and sunset periods. The transition period should be minimized by the non-synchronous Generating Facilities as much as possible.
 - b. When the Generating Facility is physically connected but operating at zero MW and zero Mvar as measured at the high-side of the generator substation, the Generating Facility is not required to control system voltage.
 - c. The requirements in bullet a and b above are minimum requirements. If a nonsynchronous Generating Facility has the capability to provide dynamic voltage control and maintain ATC's voltage schedule when it is at zero MW output, that is acceptable and preferred by ATC.

Static reactive power devices (e.g., capacitors and inductors) can only be used to make up for

- Inductive losses between the generator terminal and the POI for synchronous generators, or
- Inductive losses or collector system charging between the generator terminal and the high side of generator substation for non-synchronous generators.

All other reactive power needed to meet the power factor requirement must be provided by continuous and sustainable dynamic sources. Operation across the entire power factor range must be fully dynamic, variable, and capable of sustained indefinite operation.

Static sources can be switched on or off in the range of seconds and provide reactive power in large discrete blocks. Cap Banks are considered static sources of reactive power.

Dynamic sources can provide variable amounts of reactive power in a few milliseconds. Static Var Compensators (SVCs), Static Synchronous Compensators (STATCOMs), Flexible AC Transmission Systems (FACTS), inverters, and synchronous condensers are all considered dynamic sources of reactive power.

For non-synchronous generation projects in the DPP 2021 Cycle Phase 1 study group, if they did not have a signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

As shown in Table 1.2.2-1, all requests in this queue met FERC Order 827 and ATC Criteria for dynamic reactive power requirements.

Table 1.2.2-1 – Additional Dynamic Mvar to meet ATC Dynamic Inductive Power Factor Requirement and FERC Order 827 Power Factor Requirements

MISO Queue #	Type	Additional Dynamic Reactive Compensation (Mvar)
N/A	N/A	N/A

The generation requests shown in Table 1.2.2-2 did not meet the reactive power requirements per ATC Criteria at the time of the model completion and are required to provide additional reactive power sources. All other requests in this queue met FERC Order 827 and ATC Criteria for reactive power requirements.

Table 1.2.2-2 – Additional Mvar to meet ATC Capacitive Power Factor and FERC Order 827 Power Factor Requirements

MISO Queue #	Type	Additional Shunt Compensation ^{1,2} (Mvar)
J1931	Wind	2.0
J1934	Battery Storage	6.6
J1935	Solar	1.4
J1943	Battery Storage	6.0

J1993	Solar	9.1
J2099	Solar	53.2
J2107	Solar	53.2
J2185	Solar	53.2
J2193	Solar/Battery	5.3
J2304	Battery Storage	13.2

¹ Additional compensation is required to meet the criteria at the POI Bus for synchronous Generating Facilities or the high-side of the generator substation for asynchronous Generating Facilities.

² FERC Order 827 allows this additional reactive power to be supplied by static resources, such as mechanically switched capacitors. ATC recommends that this additional reactive power be supplied by dynamic resources.

When non-synchronous Generating Facility is not generating active power (i.e. zero MW output), the reactive power injection to the transmission system at the high-side of the generator substation should be zero Mvar. Table 1.2.2-2 shows the Mvar levels expected at the high-side of the generator substations for all non-synchronous generation requests when not generating. Inductive power devices or the generating facilities inverters can be used to counter var flows created at low or no output levels.

Table 1.2.2-3 – Assessment of Reactive Power Injection to the Transmission System When Non-synchronous Generator is at zero MW output

MISO Queue #	Additional inductive Mvar required at the high side of generator substation to meet ATC Power Factor Requirement when non-synchronous generator is at zero MW output
J1895	8.9
J1931	0.7
J1934	1.3
J1935	0.9
J1943	3.4
J1993	0.1
J2029	0.4
J2055	2.2
J2099	4.0
J2107	4.0
J2148	0.0
J2185	4.0
J2193	1.6
J2211	10.0
J2304	3.2
J2316	0.5

1.2.3 Island Detection and Operation

In circumstances where the Generating Facility lack adequate governor controls such that the transmission system could result in a situation where generation could be islanded with load for the outage of two concurrent transmission elements, ATC will require the Customer to curtail their generation to zero MW gross output after the first element outage, unless operating steps can be developed to avoid the creation of such an island for the next contingency. Alternatively, upon ATC's review and approval, the Customer may choose to fund the design and installation of an automated Anti-Islanding Scheme (AIS) on the relevant transmission lines to eliminate the need of generation curtailment for a single outage event.

Based on the system configuration studied in this DPP cycle, the following Generating Facilities that lack adequate governor controls would result in an island with load for the outage of two transmission circuits (i.e. a NERC Category P6 event).

- J1934

Beyond NERC Planning Events, real-time prior outage conditions could also lead to a potential islanding condition for the next contingency. To operate the system reliably, ATC System Control Operators will take actions necessary to protect safety and reliability, up to disconnection of the power plant.

1.3 Total Network Upgrades for all Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Energy Resource Interconnection Service and Network Resource Interconnection Service as of the SIS report date. The total cost of Network Upgrades required for each generator interconnection request is listed in Table 1.3-1. The costs for Network Upgrades are planning level estimates and subject to be revised in the facility studies. All Interconnection Facility Project Diagrams are documented in Appendix C and all Network Upgrade Project Diagrams are documented in Appendix D (No project diagrams are developed for line upgrades).

Table 1.3-1 – Total Cost of Network Upgrades for DPP 21 Cycle Phase 3 Generator Interconnection Requests

MISO Queue #	Requested MW	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrade (\$)	Total Cost of Network Upgrades (Exclude TOIF & Affected Systems) (\$)	M2 Received (\$) ¹	M3 Received (\$) ²	M4 Received (\$) ³
		Steady - State Thermal & Voltage	Transient Stability	Short Circuit	Affected System		TO Network Upgrades	TO-Owned Direct Assigned (TOIF) ⁴					
[a]	[b]	[c]	[d]	[e]	[f]	[g]	[h]	[i]	[j]	[k]=[c]+[d]+[e]+[g]+[h]+[j]	[l] = \$4000 x [b]	[m]= (10%of [k])- [l]	[n]= (20%of [k])- [l] –[m]
J1895	200	14,666,420	0	0	209,985	749,673	4,923,000	581,000	0	20,339,093	800,000	1,253,594	31,572
J1931	40	0	0	0	0	47,703	25,000	0	0	72,703	160,000	0	0
J1934	150	0	0	0	141,684	144,793	5,755,000	1,534,000	0	5,899,793	600,000	0	1,361,874
J1935	50	0	0	0	0	3,836,043	940,000	855,000	0	4,776,043	200,000	0	535,380
J1943	200	0	0	0	25,028	16,382,862	225,000	0	0	16,607,862	800,000	159,268	3,457,535
J1993	99	0	0	0	35,059	10,473,718	225,000	0	0	10,698,718	396,000	0	1,512,421
J2029	62.5	0	0	0	0	8,884,115	2,602,000	793,000	0	11,486,115	250,000	398,395	1,888,994
J2055	100	0	0	0	345,144	17,898,211	4,538,000	0	0	22,436,211	400,000	185,641	2,415,757
J2099	399.03	0	0	0	116,475	21,360,838	9,164,000	425,000	0	30,524,838	1,596,120	1,056,396	18,353,243
J2107	399.03	0	0	0	116,475	21,360,838	9,164,000	425,000	0	30,524,838	1,596,120	1,056,396	18,353,243
J2148	150	0	0	0	160,175	20,921,510	1,618,000	1,311,000	0	22,539,510	800,000	81,565	3,604,077
J2185	399.03	0	0	0	116,475	21,360,838	9,164,000	425,000	0	30,524,838	1,596,120	1,056,396	18,353,243
J2193	100	0	0	0	0	10,641,873	8,862,000	927,000	0	19,503,873	800,000	1,662,627	1,404,447
J2211	200	60,896,506	0	0	854,121	30,680,387	1,304,000	1,052,000	0	92,880,893	800,000	163,538	4,826,049
J2304	400	0	0	0	37,379	30,440,620	31,578,900	1,411,000	0	62,019,520	1,600,000	536,004	9,054,555
J2316	77.5	0	0	0	0	10,179,541	25,000	0	0	10,204,541	310,000	351,933	2,653,665
Total (\$)	-	75,562,926	0	0	2,158,000	225,363,563	90,112,900	9,739,000	0	391,039,389	12,704,360	9,298,536	87,806,055

¹ M2: Milestone Payment dollars received by MISO.

² M3 = Milestone Payment dollars received by MISO.

³ M4= (20% of NU)-M2-M3.

⁴ TOIF: Transmission Owner shall collect from Interconnection Customer a tax gross-up amount on the payments made to Transmission Owner using the Transmission Owner rate in effect at the time the payment is received from Interconnection Customer.

⁵ ATC projects are estimated in ISD dollars with a 3.5% annual escalation rate and include a contingency based on a project risk generally between 5% – 20%.

1.4 Stability Operating Restrictions Summary

For NERC Category P6 stability constraints, operating restrictions may be required under prior outage conditions to avoid instability caused by the next event. A summary of all P6 stability operating restrictions (maximum allowed real power outputs) identified for study generators in the DPP 2021 Cycle 1 Phase 3 are provided in Table 1.4-1. Given the expectation that plant models will be changing with generator modifications and as-built parameters, P6 stability operating restrictions were not calculated as part of this DPP study and will be performed at a later stage. These constraints are discussed in further detail in Sections 3 and 6.

Table 1.4-1 – NERC Category P6 Stability Operating Restriction Summary

MISO Queue #	Prior Outage	Faulted Element
J1931		
J2193		
J1895		
J2099, J2107, J2185		

1.5 In-Service Dates and Cost Estimates

ATC understands that the estimated in-service date may not align with the Interconnection Customer's Synchronization Date; however, negotiated and executed agreements, such as an Engineering and Procurement Agreement, can be used prior to the GIA execution date to expedite Network Upgrades. In absence of any special arrangement, typical times to develop a new Interconnection Facility is about 24-48 months after the GIA is executed, assuming no delays due to Interconnection Customer's permits, state processes, land acquisitions, deliverables (such as a finish graded substation site, etc.) It also assumes that system outages required to construct facilities can be obtained timely.

1.6 MTEP Projects

If a MTEP transmission project(s) resolves the constraint, and that project(s) is approved by the Board within (1) calendar year of the GIA execution or execution of an amendment thereof, then the Interconnection Customer will not be responsible for transmission upgrade(s) that would resolve the constraint. If that MTEP project(s) is not approved within one (1) calendar year of the GIA execution or execution of an amendment thereof, the Interconnection Customer will be responsible for those transmission upgrade(s).

1.7 Further Study

The next step in the MISO Generator Interconnection Procedures is to perform additional SISs (if needed), Interconnection Customer Interconnection Facility Studies, and Network Upgrade Facility Studies. Those Facilities Studies will specify in more detail the time and cost of the equipment, engineering, procurement, and construction of the Interconnection Facilities and Network Upgrades identified in this report.

1.8 Compliance Summary

This study report partially meets NERC TPL-001-4 standard, FAC-002-3 standard, and Local Planning Criteria. In ATC's annual Ten-Year Assessment (TYA) and MISO annual MTEP studies, additional compliance related studies will be performed for the generator interconnection requests with signed GIAs. Appendix J describes in detail the NERC and Local Criteria requirements met by this SIS report.

2.0 STEADY-STATE ANALYSIS

Steady-state analysis was performed to identify thermal and voltage upgrades required to interconnect the generator interconnection requests in the DPP 2021 Cycle Phase 3 to the transmission system. Detailed study assumptions, criteria, and methodology are documented in Appendix A.

2.1 Model Development

2.1.1 Study Cases

Three study cases for the steady-state thermal and voltage analysis were developed based on MTEP21 Target A models built for 2026: 2026 summer peak batteries discharging, 2026 shoulder batteries discharging, and 2026 shoulder batteries charging. The ATC system in these cases was updated with the most recent data available at the time of model construction. The cases were reviewed by ATC and the Interconnection Customers. Based on this review, the cases were further modified to account for model updates, changes, and competing generation requests that had dropped out of the MISO queue since the models were built.

MISO Long Range Transmission Planning (LRTP) Tranche 1 projects 4 and 6 were identified as ERIS Network Upgrades in the MISO DPP 2020 Cycle 1 East

(ATC) study. LRTP Tranche 1 projects 4 and 6 were included in the study cases before the ERIS and NRIS analysis.

MISO LRTP Tranche 1 projects 2, 7, and 8 were also added to the study cases due to them being identified as voltage Network Upgrades in DPP 2021 West area study. All LRTP Tranche 1 projects were modeled using MTEP project definitions as of 8/10/2023.

The prior queued generator interconnection requests in the ATC system that are in the DPP process at the time of the model development are included in the study cases. Associated Network Upgrades for projects prior to DPP-20 cycle were also included.

Public information related to the MISO Generator Interconnection Request queue can be found at:

https://www.misoenergy.org/planning/resource-utilization/GI_Queue/

The summer peak batteries discharging, shoulder batteries discharging, and shoulder batteries charging cases dispatched generation within MISO according to section 6.1.1.1.2, Study Case Development, in the MISO BPM-015-r23.

All excess generation from this methodology is dispatched against all units in MISO Classic proportionally, excluding the units in the current DPP cycle. Scheduled firm transfers are ignored in this dispatch methodology.

2.1.2 Benchmark Cases

Two benchmark cases were used to benchmark system performance without the DPP 2021 Cycle Phase 3 generating facilities and were created by taking the DPP 2021 Cycle Phase 3 Generating Facilities offline from the corresponding study cases. The MISO Classic area was used for power balance, where generation was scaled in proportion to P_{max} minus P_{gen} .

2.2 Reactive Power Requirements (FERC Order 827)

All synchronous and non-synchronous generation in this queue were evaluated to determine if the requests meet FERC Order 827 and ATC Planning Criteria. Refer to PLG-METH-0005 in Appendix B for details on ATC's power factor analysis methodology. All of the reactive resources modeled in the assessment are summarized in Table 2.2-1.

Table 2.2-1 – Reactive Resources Modeled in Generator Interconnection Power Factor Analysis

MISO Queue #	Machines				Dynamic Devices			Additional Devices		
	Description	Real Power Pmax (MW)	Capacitive Reactive Power Qmax (Mvar)	Inductive Reactive Power Qmin (Mvar)	Description	Capacitive Reactive Power (Mvar)	Inductive Reactive Power (Mvar)	Description ¹	Capacitive Reactive Power (Mvar)	Inductive Reactive Power (Mvar)
J1895	Battery Storage	213.548	106.610	-106.61	None	N/A	N/A	None	N/A	N/A
J1931	Wind Generation	45.12	21.852	-21.852	None	N/A	N/A	None	N/A	N/A
J1934	Battery Storage	154.98	75.06	-75.06	None	N/A	N/A	None	N/A	N/A
J1935	Solar Generation	52.92	25.630	-25.63	None	N/A	N/A	None	N/A	N/A
J1943	Solar Generation	204.12	98.86	-98.86	None	N/A	N/A	2X4 Mvar Cap	8	N/A
J1993	Solar Generation	103.04	56.680	-56.68	None	N/A	N/A	1X5 Mvar Cap	35	N/A
J2029	Solar Generation	63.9	40.824	-40.824	None	N/A	N/A	None	N/A	N/A
J2055	Solar Generation	102.66	66.215	-46.35	None	N/A	N/A	None	N/A	N/A
J2099	Solar Generation	404.7057	133.020	-133.0203	None	N/A	N/A	None	N/A	N/A
J2107	Solar Generation	404.7057	133.020	-133.0203	None	N/A	N/A	None	N/A	N/A
J2148	Solar Generation	153.2	95.728	-95.728	None	N/A	N/A	None	N/A	N/A
J2185	Solar Generation	404.705	133.020	-133.0203	None	N/A	N/A	None	N/A	N/A
J2193	Solar Generation	100.0	40.769	-40.769	None	N/A	N/A	None	N/A	N/A
	Battery Storage	25.0	10.872	10.872	None	N/A	N/A			
J2211	Battery Storage	216.688	108.179	-108.179	None	N/A	N/A	None	N/A	N/A
J2304	Battery Storage	406.262	224.784	-224.784	None	N/A	N/A	None	N/A	N/A
J2316	Solar Generation Addition	86.5	41.900	-41.9	None	N/A	N/A	None	N/A	N/A

¹ FERC Order 827 allows this additional reactive power to be supplied by static resources, such as mechanically switched capacitors. ATC recommends that this additional reactive power be supplied by dynamic resources.

The dynamic capacitive power factor requirement analysis showed all requests meeting ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-2.

Table 2.2-2 – Assessment of Dynamic Capacitive Power Factor Requirement

MISO Queue #	Machine Terminal Bus #	Capability at Machine Terminal		Additional Dynamic Capacitive Reactive Power ¹ (Mvar)	Dynamic Power Factor Provided	Meets Requirement ² ?	Additional Requirement ³ (Mvar)
		Real Power Pmax (MW)	Capacitive Reactive Power Qmax (Mvar)				
J1895	48950	213.5	106.600	0.0	0.89	Yes	0.0
J1931	49310	45.1	21.900	0.0	0.90	Yes	0.0
J1934	49340	154.98	75.06	0.0	0.90	Yes	0.0
J1935	49350	52.9	25.630	0.0	0.90	Yes	0.0
J1943	49430, 49436	204.2	98.800	0.0	0.90	Yes	0.0
J1993	49930	103.0	56.679	0.0	0.88	Yes	0.0
J2029	50290, 50291	63.9	40.824	0.0	0.84	Yes	0.0
J2055	50550	102.1	66.2	0.0	0.84	Yes	0.0
J2099	50990, 50991	404.8	133.018	0.0	0.95	Yes	0.0
J2107	51070, 51071	404.8	133.018	0.0	0.95	Yes	0.0
J2148	51480	153.66	95.728	0.0	0.85	Yes	0.0
J2185	51850, 51851	404.6	133.018	0.0	0.95	Yes	0.0
J2193	51930, 51931	125.0	51.641	0.0	0.92	Yes	0.0
	51930 only	100.0	40.769	0.0	0.93	Yes	0.0
	51931 only	25.0	10.872	0.0	0.92	Yes	0.0
J2211	52110	217.6	108.2	0.0	0.89	Yes	0.0
J2304	53040, 53041	406.2	224.8	0.0	0.87	Yes	0.0
J2316	53160	85.5	41.900	0.0	0.90	Yes	0.0

¹ Dynamic capacitive reactive power provided by Interconnection Customer owned equipment in addition to the machine.

² ATC requires a 0.95 ATC Capacitive Dynamic Power Factor.

³ Additional dynamic reactive power required to meet ATC Capacitive Dynamic Power Factor.

⁴ Gross values. The generating facility auxiliary load were not included.

The additional capacitive power factor requirement analysis showed that 10 requests do not meet the ATC Criteria or FERC Order 827 requirements. The results are summarized in Table 2.2-3.

Table 2.2-3 – Assessment of Additional Capacitive Power Factor Requirement

MISO Queue #	Point of Measurement ¹	Capability at Point of Measurement ¹		Power Factor	Meets Requirement?	Additional Requirement (Mvar)
		Real Power (MW)	Reactive Power (Mvar)			
J1895	48953	209.7	68.6	0.95	Yes	0.0
J1931	44833	44.6	12.7	0.96	No	2.0
J1934	49343	152.6	43.6	0.96	No	6.6
J1935	49353	52.3	15.8	0.96	No	1.4
J1943	49430, 49436	201.4	60.2	0.96	No	6.0
J1993	46293	99.6	23.6	0.97	No	9.1
J2029	50296	62.9	26.7	0.92	Yes	0.0
J2055	693319	100.0	41.7	0.92	Yes	0.0
J2099	50996	400.5	78.4	0.98	No	53.2
J2107	50996	400.5	78.4	0.98	No	53.2
J2148	51480	151.3	60.3	0.93	Yes	0.0
J2185	50996	400.5	78.4	0.98	No	53.2
J2193	51935	123.9	35.4	0.96	No	5.3
	Solar Only	99.1	31.6	0.95	Yes	0.0
	Storage Only	24.9	10.9	0.92	Yes	0.0
J2211	52113	210.8	69.2	0.95	Yes	0.0
J2304	53046	400.6	118.5	0.96	No	13.2
J2316	48143	85.5	27.9	0.95	Yes	0.0

¹ Point of Measurement is the POI Bus for synchronous machines and high side of generator substation for asynchronous machines.

The dynamic inductive power factor requirement analysis showed all of the requests meet ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-4.

Table 2.2-4 – Assessment of Dynamic Inductive Power Factor Requirement

MISO Queue #	Machine Terminal Bus #	Capability at Machine Terminal		Additional Dynamic Inductive Reactive Power ¹ (Mvar)	Dynamic Power Factor Provided	Meets Requirement ² ?	Additional Requirement ³ (Mvar)
		Real Power Pmax (MW)	Inductive Reactive Power Qmin (Mvar)				
J1895	48950	213.5	-106.6	0.0	0.89	Yes	0.0
J1931	49310	45.1	-21.9	0.0	0.90	Yes	0.0
J1934	49340	154.98	-75.06	0.0	0.90	Yes	0.0
J1935	49350	52.9	-25.6	0.0	0.90	Yes	0.0
J1943	49430, 49436	204.2	-98.800	0.0	0.90	Yes	0.0
J1993	49930	103.0	-56.679	0.0	0.88	Yes	0.0
J2029	50290, 50291	63.9	-40.824	0.0	0.84	Yes	0.0
J2055	50550	102.3	-46.4	0.0	0.91	Yes	0.0
J2099	50990, 50991	404.8	-133.018	0.0	0.95	Yes	0.0
J2107	51070, 51071	404.6	-133.018	0.0	0.95	Yes	0.0
J2148	51480	153.2	-95.728	0.0	0.85	Yes	0.0
J2185	51850, 51851	404.6	-133.018	0.0	0.95	Yes	0.0
J2193	51930, 51931	125.0	-51.641	0.0	0.92	Yes	0.0
	51930 only	100.0	-40.769	0.0	0.93	Yes	0.0
	51931 only	25.0	-10.872	0.0	0.92	Yes	0.0
J2211	52110	216.7	-108.2	0.0	0.89	Yes	0.0
J2304	53040, 53041	406.4	-224.8	0.0	0.88	Yes	0.0
J2316	53160	86.5	-41.9	0.0	0.90	Yes	0.0

¹ Dynamic inductive reactive power provided by Interconnection Customer owned equipment in addition to the machine.

² ATC requires a 0.95 ATC Inductive Dynamic Power Factor.

³ Additional dynamic reactive power required to meet ATC Inductive Dynamic Power Factor.

⁴ Gross Values. The generating facility auxiliary load were not included.

The static inductive power factor requirement analysis showed all requests meeting ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-5.

Table 2.2-5 – Assessment of Static Inductive Power Factor Requirement

MISO Queue #	POI Bus (synchronous) or HV Bus (asynchronous)	MW at POI (synchronous) or HV Bus (asynchronous)	Mvar at POI (synchronous) or HV Bus (asynchronous)	POI (synchronous) or HV Bus (asynchronous) Power Factor	Meets 0.95 ATC Inductive Power Factor Requirement at POI Bus (synchronous) or HV Bus (asynchronous)?
J1895	48953	209.0	-152.3	0.81	Yes
J1931	44833	44.5	-32.8	0.80	Yes
J1934	49343	152.2	-111.9	0.81	Yes
J1935	49353	52.2	-36.8	0.82	Yes
J1943	44119	201.0	-143.6	0.81	Yes
J1993	46293	98.2	-103.3	0.69	Yes
J2029	50296	62.6	-56.5	0.74	Yes
J2055	693319	100.0	-73.4	0.81	Yes
J2099	50996	400.2	-188.0	0.91	Yes
J2107	50996	400.2	-188.0	0.91	Yes
J2148	50743	150.7	-143.3	0.72	Yes
J2185	50996	400.2	-188.0	0.91	Yes
J2193	51935	123.8	-67.7	0.88	Yes
	Solar Only	99.0	-51.2	0.89	Yes
	Storage Only	24.9	-10.5	0.92	Yes
J2211	52113	209.8	-155.6	0.80	Yes
J2304	53046	400.1	-340.0	0.76	Yes
J2316	48143	85.5	-56.9	0.83	Yes

When a non-synchronous Generating Facility is not generating active power (i.e. zero MW output), the reactive power injection to the transmission system at the high-side of the generator substation should be zero Mvar. Table 2.2-6 shows the Mvar levels expected at the high-side of the generator substations for all non-synchronous generation requests when not generating. Inductive power devices or the generating facilities inverters can be used to counter var flows created at low or no output levels.

Table 2.2-6 – Assessment of Reactive Power Injection to the Transmission System When Non-synchronous Generator is at zero MW output

MISO Queue #	Additional inductive Mvar required at the high side of generator substation to meet ATC Power Factor Requirement when non-synchronous generator is at zero MW output
J1895	8.9
J1931	0.7
J1934	1.3
J1935	0.9
J1943	3.4
J1993	0.1
J2029	0.4
J2055	2.2
J2099	4.0
J2107	4.0
J2148	0.0
J2185	4.0
J2193	1.6
J2211	10.0
J2304	3.2
J2316	0.5

2.3 NERC TPL Contingency Analysis Results

The incremental impact of the proposed generator interconnection on transmission facilities was evaluated by comparing steady state power flows and voltages between benchmark cases (without DPP 2021 Cycle Phase 3 projects) and study cases (with DPP 2021 Cycle Phase 3 projects). Post-contingency cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment enabled, and switched shunt adjustment enabled. Detailed NERC TPL Category P contingencies that were studied are described in Table A.2.1-1 in Appendix A.

2.3.1 2026 Summer Peak Batteries Discharging

The study identified the steady-state thermal constraints that qualified as MISO Injection Constraints in the 2026 Summer Peak Discharging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2026 Summer Peak Discharging MISO Injection Constraints that

require Network Upgrades is presented in Table 2.3.1. No voltage constraints were identified in the steady-state analyses.

Table 2.3.1 – 2026 Summer Peak Batteries Discharging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by NERC TPL Planning Events	Responsible Generator(s)
2026 Summer Peak Batteries Discharging	Rocky Run – J2099 POI 345 kV line	ATC	Base Case, P1.2, P2.3, P7.1	J2029 J2055 J2099 J2107 J2148 J2185 J2211 J2316
	J2099 POI – Werner West 345 kV line	ATC	P1.2, P2.3, P7.1	J2099 J2107 J2185
	Eau Claire 345/161 kV transformer T10	XEL	P1.2	J2099 J2107 J2185

2.3.2 2026 Shoulder Batteries Discharging

The study identified the steady-state thermal constraints that qualified as MISO Injection Constraints in the 2026 Shoulder Discharging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2026 Shoulder Batteries Discharging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.2. No voltage constraints were identified in the steady state analyses.

Table 2.3.2 – 2026 Shoulder Batteries Discharging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by NERC TPL Planning Events	Responsible Generator(s)
None				

2.3.3 2026 Shoulder Batteries Charging

The study identified the steady-state thermal constraints that qualified as MISO Injection Constraints in the 2026 Shoulder Batteries Charging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2026 Shoulder Batteries Charging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.3. No voltage constraints were identified in the steady state analyses.

Table 2.3.3 – 2026 Shoulder Batteries Charging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by NERC TPL Planning Events	Responsible Generator(s)
2026 Shoulder Batteries Charging	Rocky Run – J2099 POI 345 kV line	ATC	P7.1	J2211
	J2099 POI – Werner West 345 kV line	ATC	P7.1	J2211
	North Appleton – Lost Dauphin 138 kV line	ATC	Base Case, P2.3, P2.4, P7.1	J2211
	Pulliam – Tower Dr 138 kV line	ATC	P1.2, P2.2, P2.3	J2211
	Packaging – Pulliam 138 kV line	ATC	P1.2, P2.2, P2.3, P2.4	J2211
	Tower Dr – Preble 138 kV line	ATC	P5.5, P7.1	J2211
	Preble – X154 Reactor 138 kV line	ATC	P5.5, P7.1	J2211
	X154 Reactor – Highway V 138 kV line	ATC	P5.5, P7.1	J2211
	Badger – Cloverleaf 138 kV line	ATC	P2.4, P5.5, P7.1	J2211
	Cloverleaf – Maplewood 138 kV line	ATC	P2.4, P5.5, P7.1	J2211
	Brick Church – Cobblestone Tap 69 kV line	ATC	P5.5	J1895
	Cobblestone Tap – Zenda Tap 69 kV line	ATC		
	Zenda Tap – Katzenberg 69 kV line	ATC		
	Paddock – Paddock Reactor 138 kV line	ATC	P2.2, P2.4	J1895
	Paddock Reactor – Northwest Beloit 138 kV line	ATC	P2.3, P2.4	J1895
	Northwest Beloit – Blackhawk 138 kV line	ATC	P2.3, P2.4	J1895
	Blackhawk – Colley Road 138 kV line	ATC	P2.4	J1895
	North Lake Geneva – Walworth 69 kV line	ATC	P1.2, P2.2, P2.3	J1895

2.3.4 Network Upgrades Identified in ERIS Analysis

Based on the steady-state analyses, the worst loading of each facility under NERC TPL Planning Events that meets MISO Injection Constraint criteria is shown in Table 2.3.4-1. Identified Network Upgrades are also included. Good faith Cost Estimates of the ERIS thermal Network Upgrades identified in the steady-state analysis for the 2026 scenarios are listed in Table 2.3.4-2. Detailed cost allocations are provided in appendix E.

Table 2.3.4-1 – ERIS Network Upgrades Identified to Address MISO Steady-State Injection Constraints

Steady-State Injection Constraint	Responsible Generator(s)	Facility Owner	Study Case	Applicable Rating (MVA)	Worst Loading (%)	NERC TPL Planning Event	Network Upgrades
Rocky Run – J2099 POI 345 kV line	J2029 J2055 J2099 J2107 J2148 J2185 J2211 J2316	ATC	26SUM	861	162.17		MTEP PID# 50560 (LRTP-27: Rocky Run - Werner - North Appleton)
J2099 POI – Werner West 345 kV line	J2099 J2107 J2185 J2211	ATC	26SUM	861	136.59		MTEP PID# 50560 (LRTP-27: Rocky Run - Werner - North Appleton)
Eau Claire 345/161 kV transformer T10	J2099 J2107 J2185	ATC	26SUM	433.9	129.87		MTEP PID# 23372 (LRTP-05: Tremval – Eau Claire – Jump River)
Pulliam – Tower Dr 138 kV line	J2211	ATC	26SH, Batteries Charging	235	111.53		Pulliam – Tower Dr (V-100) 138 kV, Rebuild
Packaging – Pulliam 138 kV line	J2211	ATC	26SH, Batteries Charging	235	105.37		Pulliam – Packaging (S-123) 138 kV, Rebuild
Tower Dr – Preble 138 kV line	J2211	ATC	26SH, Batteries Charging	235	159.36		Tower Dr – Preble (X-154) 138 kV, Rebuild
Preble – X154 Reactor 138 kV line	J2211	ATC	26SH, Batteries Charging	235	174.32		Preble – Highway V (X-158) 138 kV, Rebuild
X154 Reactor – Highway V 138 kV line	J2211	ATC	26SH, Batteries Charging	237	172.84		
North Appleton – Lost Dauphin 138 kV line	J2211 J2218	ATC	26SH, Batteries Charging	235	108.48		North Appleton – Lost Dauphin (I-113) 138 kV, Rebuild
Badger – Cloverleaf 138 kV line	J2211	ATC	26SH, Batteries Charging	95	125.24		Badger – Cloverleaf – Maplewood (W-75) 138 kV, Uprate
Cloverleaf – Maplewood 138 kV line	J2211	ATC	26SH, Batteries Charging	87	127.33		
Paddock – Paddock Reactor 138 kV line	J1895	ATC	26SH, Batteries Charging	335	112.92		Rock River – Townline Road 138 kV, New Line
Paddock Reactor – Northwest Beloit 138 kV line	J1895	ATC	26SH, Batteries Charging	335	112.92		
Northwest Beloit – Blackhawk 138 kV line	J1895	ATC	26SH, Batteries Charging	335	111.14		
Blackhawk – Colley Road 138 kV line	J1895	ATC	26SH, Batteries Charging	335	100.97		
North Lake Geneva – Walworth 69 kV line	J1895	ATC	26SH, Batteries Charging	59	113.15		North Lake Geneva - Walworth 69 kV (Y-41), Uprate
Brick Church – Cobblestone Tap 69 kV line	J1895	ATC	26SH, Batteries Charging	78	138.5		MTEP PID# 20201 (Small Capital Project and Asset Renewal Program 2024)
Cobblestone Tap – Zenda Tap 69 kV line	J1895	ATC	26SH, Batteries Charging	69	145.66		

Zenda Tap – Katzenberg 69 kV line	J1895	ATC	26SH, Batteries Charging	69	138.55		
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Table 2.3.4-2 – ERIS Network Upgrades and Cost Estimates

Steady-State Injection Constraint	Facility Owner	Network Upgrade	Cost (\$) ^{1, 2}
Rocky Run – J2099 POI 345 kV line	ATC	MTEP PID# 50560 (LRTP-27: Rocky Run - Werner - North Appleton)	0
J2099 POI – Werner West 345 kV line	ATC	MTEP PID# 50560 (LRTP-27: Rocky Run - Werner - North Appleton)	0
Eau Claire 345/161 kV transformer T10	XEL	MTEP PID# 23372 (LRTP-05: Tremval – Eau Claire – Jump River)	0
Pulliam – Tower Dr 138 kV line	ATC	Pulliam – Tower Dr (V-100) 138 kV, Rebuild	3,837,891
Packaging – Pulliam 138 kV line	ATC	Pulliam – Packaging (S-123) 138 kV, Rebuild	3,849,069
Tower Dr – Preble 138 kV line	ATC	Tower Dr – Preble (X-154) 138 kV, Rebuild	10,378,000
Preble – X154 Reactor 138 kV line X154 Reactor – Highway V 138 kV line	ATC	Preble – Highway V (X-158) 138 kV, Rebuild	8,641,000
North Appleton – Lost Dauphin 138 kV line	ATC	North Appleton – Lost Dauphin (I-113) 138 kV, Rebuild	25,944,000
Badger – Cloverleaf 138 kV line	ATC	Badger – Cloverleaf - Maplewood (W-75) 138 kV, Uprate	8,246,546
Cloverleaf – Maplewood 138 kV line			
Paddock – Paddock Reactor 138 kV line Paddock Reactor – Northwest Beloit 138 kV line Northwest Beloit – Blackhawk 138 kV line Blackhawk – Colley Road 138 kV line	ATC	Rock River – Townline Road 138 kV, New Line	9,084,939
North Lake Geneva – Walworth 69 kV line	ATC	North Lake Geneva - Walworth 69 kV (Y-41), Uprate	5,581,481
Brick Church – Cobblestone Tap 69 kV line	ATC	MTEP PID# 20201 (Small Capital Project and Asset Renewal Program 2024)	0
Cobblestone Tap – Zenda Tap 69 kV line			
Zenda Tap – Katzenberg 69 kV line			

¹ ATC Network Upgrades included a contingency based on a project risk generally between 5% – 20%.

² All Network Upgrades were estimated on an estimated ISD assuming GIA execution by 12/31/2025

2.3.5 Network Upgrade Alternatives Considered

All ERIS network upgrades identified in Table 2.3.4-2, except for “Rock River – Townline Road 138 kV, New Line” are direct upgrades of the ERIS thermal constraint facilities to ATC design standards and considered as least-cost solutions. Therefore, no other alternatives were examined for those solutions.

For “Rock River – Townline Road 138 kV, New Line” one alternative was considered. The alternative was to rebuild a total of 7.34 miles of Paddock – Northwest Beloit – Blackhawk – Colley Rd 138 kV and remove the Paddock reactor. Compared to the new 0.45 miles Rock River – Townline Road 138 kV line, this alternative came back at a much higher cost. Therefore, ATC decided on “Rock River – Townline Road 138 kV, New Line” as the required network upgrade.

2.3.6 Potential Operating Restriction

Study generators may be subject to potential steady-state operating restrictions under prior outage conditions. Real-time steady-state thermal constraints due to

NERC Category P3/P6 events (N-1-1) will be mitigated in the day-ahead and real-time market through the MISO binding constraint and other operating procedures.

2.4 MWEX Voltage Stability Analysis

This analysis was performed on the 2026 shoulder charging study case to determine if the MISO DPP 2021 Cycle Phase 3 projects could cause or contribute to voltage stability violations on MWEX (Minnesota Wisconsin Export Interface). LRTP projects 4 and 6 were included in the case due to the projects being identified as network upgrades in the ATC region for DPP-20. The study results show that the post-DPP scenario is voltage stable with LRTP projects 4 and 6 and meets ATC planning criteria. The detailed report can be found in Appendix K.

3.0 STABILITY ANALYSIS

3.1 Model Development

Stability analysis in Section 3 was performed using TSAT V22 to evaluate the stability performance of the generating facilities in the DPP 2021 Cycle 1 Phase 3. Detailed study assumptions, criteria, and methodology are documented in Appendix A. For generators that require Electromagnetic transient (EMT) stability analysis using Power Systems Computer-Aided Design (PSCAD), results are documented in Section 6 of the report.

Stability plots consist of generator rotor angles, generator real power output, generator reactive power output, generator terminal voltages, and transmission bus voltages for each simulation. Simulations were performed with a 0.5 second flat start followed by the appropriate disturbance. Simulations were run for a 20-second duration.

3.1.1 Study Case

A stability study case representing the 2026 shoulder load condition was developed by MISO and reviewed by ATC and Interconnection Customers. The model was developed and studied in TSAT v22. TSAT, like PSSE, is a root mean squared (RMS) based program that only models a network's positive sequence quantities. All units under study were dispatched to 100% of P_{MAX}. Additionally, the following local units were also dispatched to 100% of P_{MAX} to meet ATC Planning Criteria:

Table 3.1.1-1 – Local Unit Dispatch for Stability Analysis

Queue #	Nearby Units to Dispatch to P _{MAX}
J1895	J850, Paris, J878, J1316, J1778
J1931	J1483, J1781, J855, J1374, J870, J871, J1773, J1000
J1934	None
J1935	None
J1943	Columbia, J1410, J1411, J1746, J1732
J1993	J1629, South Fondu Lac, Columbia
J2029	J1814, Mihm, Kuester
J2055	Cedar Ridge, R1014

J2099	Weston, J1573, J1513
J2107	Weston, J1573, J1513
J2148	WestonJ1573, J1513
J2185	Weston, J1573, J1513
J2193	J1002, J986, J1751
J2211	Pullum (PUL G31)
J2304	RockGen, Columbia, J1214, J1326, J1779, J1377, Riverside
J2316	J1814, Mihm, Kuester

3.1.2 Benchmark Case

The DPP 2021 Phase 3 generating facilities were removed from the study case. The MISO Classic methodology, where generation was scaled in proportion to Pmax minus Pgen, was used for power balance.

Faults were simulated using the 2026 shoulder study case. If a transient stability criteria constraint was identified, the same disturbance was repeated in the benchmark case.

3.2 J1895 Stability Study

3.2.1 J1895 Stability Results

Stability results for 1985 can be found in Appendix G. Under all faults, except two P6 events, the simulations show the system meeting all transient stability criteria.

3.2.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the 1985 generating facilities.

3.2.3 Operating Restrictions Identified in the Stability Study

For P6 events stability operating restrictions for J1895 were not calculated as part of this DPP study and will be performed at a later stage.

3.3 J1931 Stability Study

3.3.1 J1931 Stability Results

Stability results for J1931 can be found in Appendix G. Under all faults for J1931, the simulations show the system meeting all transient stability criteria.

3.3.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1931 generating facilities.

3.3.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1931 generating facilities.

3.4 J1934 Stability Study

3.4.1 J1934 Stability Results

Stability results for J1934 can be found in Appendix G. Under all faults for J1934, the simulations show the system meeting all transient stability criteria.

3.4.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1934 generating facilities.

3.4.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1934 generating facilities.

3.5 J1935 Stability Study

3.5.1 J1935 Stability Results

Stability results for J1935 can be found in Appendix G. Under all faults for J1935, the simulations show the system meeting all transient stability criteria.

3.5.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1935 generating facilities.

3.5.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1935 generating facilities.

3.6 J1943 Stability Study

3.6.1 J1943 Stability Results

Stability results for J1943 can be found in Appendix G. Under all faults for J1943, the simulations show the system meeting all transient stability criteria.

3.6.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1943 generating facilities.

3.6.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1943 generating facilities.

3.7 J1993 Stability Study

3.7.1 J1993 Stability Results

Stability results for J1993 can be found in Appendix G. Under all faults for J1993, the simulations show the system meeting all transient stability criteria.

3.7.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1993 generating facilities.

3.7.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1993 generating facilities.

3.8 J2029 Stability Study

3.8.1 J2029 Stability Results

Stability results for J2029 can be found in Appendix G. Under all faults for J2029, the simulations show the system meeting all transient stability criteria.

3.8.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J2029 generating facilities.

3.8.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J2029 generating facilities.

3.9 J2055 Stability Study

3.9.1 J2055 Stability Results

Stability results for J2055 can be found in Appendix G. Under all faults for J2055, the simulations show the system meeting all transient stability criteria.

3.9.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J2055 generating facilities.

3.9.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J2055 generating facilities.

3.10 J2099 Stability Study

3.10.1 J2099 Stability Results

Stability results for J2099 can be found in Appendix G. Under all faults for J2099, the simulations show the system meeting all transient stability criteria.

3.10.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J2099 generating facilities.

3.10.3 Operating Restrictions Identified in the Stability Study

For P6 events stability operating restrictions for J2099 were not calculated as part of this DPP study and will be performed at a later stage.

3.11 J2107 Stability Study

3.11.1 J2107 Stability Results

Stability results for J2107 can be found in Appendix G. Under all faults for J2107, the simulations show the system meeting all transient stability criteria.

3.11.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J2107 generating facilities.

3.11.3 Operating Restrictions Identified in the Stability Study

For P6 events stability operating restrictions for J2107 were not calculated as part of this DPP study and will be performed at a later stage.

3.12 J2148 Stability Study

3.12.1 J2148 Stability Results

Stability results for J2148 can be found in Appendix G. Under all faults for J2148, the simulations show the system meeting all transient stability criteria.

3.12.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J2148 generating facilities.

3.12.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J2148 generating facilities.

3.13 J2185 Stability Study

3.13.1 J2185 Stability Results

Stability results for J2185 can be found in Appendix G. Under all faults for J2185, the simulations show the system meeting all transient stability criteria.

3.13.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J2185 generating facilities.

3.13.3 Operating Restrictions Identified in the Stability Study

For P6 events stability operating restrictions for J2185 were not calculated as part of this DPP study and will be performed at a later stage.

3.14 J2193 Stability Study

3.14.1 J2193 Stability Results

Stability results for J2193 can be found in Appendix G. Under all faults for J2193, the simulations show the system meeting all transient stability criteria.

3.14.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J2193 generating facilities.

3.14.3 Operating Restrictions Identified in the Stability Study

For P6 events stability operating restrictions for J2193 were not calculated as part of this DPP study and will be performed at a later stage.

3.15 J2211 Stability Study

3.15.1 J2211 Stability Results

Stability results for J2211 can be found in Appendix G. Under all faults for J2211, the simulations show the system meeting all transient stability criteria.

3.15.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J2211 generating facilities.

3.15.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J2211 generating facilities.

3.16 J2304 Stability Study

3.16.1 J2304 Stability Results

Stability results for J2304 can be found in Appendix G. Under all faults for J2304, the simulations show the system meeting all transient stability criteria.

3.16.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J2304 generating facilities.

3.16.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J2304 generating facilities.

3.17 J2316 Stability Study

3.17.1 J2316 Stability Results

Stability results for J2316 can be found in Appendix G. Under all faults for J2316, the simulations show the system meeting all transient stability criteria.

3.17.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J2316 generating facilities.

3.17.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J2316 generating facilities.

4.0 SHORT CIRCUIT ANALYSIS

Short Circuit analysis was performed in Phase 3 with the final ERIS and NRIS network upgrade projects included in ATC Protection CAPE (Computer-Aided Protection Engineering) models, which include all ATC-owned substations and all ATC local distribution customer owned substations. The analysis was performed according to ATC short circuit analysis methodology as described in Appendix A. Short circuit analysis results for facilities under MISO Functional Control are described in this section. The short circuit impact to facilities that are not under MISO Functional Control are described in Section 7.

4.1 Short Circuit Analysis Results

Maximum and minimum fault duty was calculated at the POI for each generator request and results are summarized in Table 4.1-1, Table 4.1-2, Table 4.1-3 and Table 4.1-4.

Table 4.1-1 – Outage Assumptions for Minimum Fault Duty Calculations

MISO Queue #	Contingency
J1895	
J1931	
J1934	
J1935	
J1943	
J1993	
J2029	
J2055	
J2099	
J2107	
J2148	
J2185	
J2193	
J2211	
J2304	
J2316	

Table 4.1-2 – Maximum and Minimum Fault Duty at POIs

Location	Maximum Fault Duty (Amps)		Minimum Fault Duty (Amps)	
	Three-Phase	Single-Phase	Three-Phase	Single-Phase
J1895 POI Without J1895	9460	7110	4484	3606
J1895 POI With J1895	11781	11592	4484	5307
J1931 POI Without J1931	13139	13451	6200	6457
J1931 POI With J1931	13239	13718	6200	6540
J1934 POI Without J1934	16165	11975	5098	3861
J1934 POI With J1934	16952	14878	5098	5415
J1935 POI Without J1935	21664	20603	8043	4715
J1935 POI With J1935	22182	22058	8043	6748
J1943 POI Without J1943	20638	19905	11764	12289
J1943 POI With J1943	21093	21045	11764	12753
J1993 POI Without J1993	14697	11438	4478	4181
J1993 POI With J1993	14961	12338	4478	4553
J2029 POI Without J2029	5566	3754	3306	2316
J2029 POI With J2029	5972	5112	3306	3146
J2055 POI Without J2055	13693	14489	11038	9054
J2055 POI With J2055	13887	14695	11038	9054
J2099 POI Without J2099	18110	18780	5139	6307
J2099 POI With J2099	18928	20557	5139	6565
J2107 POI Without J2107	18110	18780	5139	6307
J2107 POI With J2107	18928	20557	5139	6565
J2148 POI Without J2148	21933	20205	11119	8159
J2148 POI With J2148	22687	22038	11119	9817
J2185 POI Without J2185	18110	18780	5139	6307
J2185 POI With J2185	18928	20557	5139	6565
J2193 POI Without J2193	7158	5693	3439	3190
J2193 POI With J2193	7882	7961	3439	4091
J2211 POI Without J2211	23626	17890	19399	14066
J2211 POI With J2211	24648	22215	19399	17579
J2304 POI Without J2304	14482	10500	5849	4380
J2304 POI With J2304	15521	13265	5849	5848
J2316 POI Without J2316	6958	7024	2874	2351
J2316 POI With J2316	7489	7561	2874	3362

Table 4.1-3 – Maximum Fault Thevenin Equivalent Data

Location	Positive Sequence (Ohm)	Negative Sequence (Ohm)	Zero Sequence (Ohm)	Three-Phase Thevenin X/R Ratio	Single-Phase Thevenin X/R Ratio	Three-Phase ANSI 37.010 X/R Ratio	Single-Phase ANSI 37.010 X/R Ratio
J1895 POI Without J1895	1.2054 + j8.3355	1.2137 + j8.3364	3.0965 + j16.4892	6.92	6.01	7.14	8.71
J1895 POI With J1895	0.7216 + j6.4740	0.7265 + j6.4749	0.6565 + j7.5633	8.97	9.75	7.14	9.19
J1931 POI Without J1931	1.1816 + j15.4020	1.1962 + j15.4172	1.7249 + j17.6108	13.04	11.8	10.87	12.22
J1931 POI With J1931	1.1816 + j15.4020	1.1962 + j15.4172	1.6100 + j17.0191	13.04	12	10.87	12.24
J1934 POI Without J1934	0.5666 + j4.9214	0.5834 + j4.9220	2.1684 + j9.8384	8.69	5.93	9.95	6.23
J1934 POI With J1934	0.5666 + j4.9214	0.5834 + j4.9220	1.0340 + j6.8686	8.69	7.65	9.95	11.96
J1935 POI Without J1935	0.1930 + j1.8289	0.1969 + j1.8341	0.1810 + j2.1099	9.48	10.11	22.94	20.79
J1935 POI With J1935	0.1930 + j1.8289	0.1969 + j1.8341	0.1421 + j1.8596	9.48	10.38	22.94	21.29
J1943 POI Without J1943	0.7651 + j9.9897	0.8959 + j9.9882	1.5890 + j11.2393	13.06	9.61	17.72	14.36
J1943 POI With J1943	0.7651 + j9.9897	0.8959 + j9.9882	1.3109 + j10.2116	13.06	10.16	17.72	15.01
J1993 POI Without J1993	0.7824 + j13.0715	0.8689 + j13.1262	3.2685 + j25.8169	16.71	10.57	17.78	15.58
J1993 POI With J1993	0.8863 + j13.9873	0.9860 + j14.0492	2.5567 + j22.8609	15.78	11.49	17.78	16.9
J2029 POI Without J2029	2.4269 + j14.3670	2.5466 + j14.4423	8.5463 + j34.0704	5.92	4.65	6.67	5.04
J2029 POI With J2029	2.4269 + j14.3670	2.5466 + j14.4423	3.4190 + j21.2541	5.92	5.97	6.67	7.54
J2055 POI Without J2055	0.9573 + j14.5002	0.9904 + j14.5002	0.7773 + j12.1524	15.15	15.1	27.48	29.42
J2055 POI With J2055	0.9573 + j14.5002	0.9904 + j14.5002	0.7773 + j12.1523	15.15	15.1	27.48	29.42
J2099 POI Without J2099	1.0026 + j12.2599	1.0467 + j12.2653	0.9413 + j10.7571	12.23	11.8	13.33	13.39
J2099 POI With J2099	1.0026 + j12.2599	1.0467 + j12.2653	0.8307 + j9.1726	12.23	11.7	13.33	12.97
J2107 POI Without J2107	1.0026 + j12.2599	1.0467 + j12.2653	0.9413 + j10.7571	12.23	11.8	13.33	13.39
J2107 POI With J2107	1.0026 + j12.2599	1.0467 + j12.2653	0.8307 + j9.1726	12.23	11.7	13.33	12.97
J2148 POI Without J2148	0.2645 + j3.6598	0.2702 + j3.6563	0.5142 + j4.7264	13.83	11.48	23.98	21.25
J2148 POI With J2148	0.2645 + j3.6598	0.2702 + j3.6563	0.4078 + j4.1047	13.83	12.12	23.98	21.4
J2185 POI Without J2185	1.0026 + j12.2599	1.0467 + j12.2653	0.9413 + j10.7571	12.23	11.8	13.33	13.39
J2185 POI With J2185	1.0026 + j12.2599	1.0467 + j12.2653	0.8307 + j9.1726	12.23	11.7	13.33	12.97
J2193 POI Without J2193	2.7983 + j11.3224	2.7996 + j11.3250	4.2458 + j20.3650	4.05	4.37	4.31	6.06
J2193 POI With J2193	2.7983 + j11.3224	2.7996 + j11.3250	1.3487 + j11.3687	4.05	4.9	4.31	5.75
J2211 POI Without J2211	0.4631 + j3.3403	0.4837 + j3.3316	1.1424 + j6.5242	7.21	6.32	12.22	11.66
J2211 POI With J2211	0.4631 + j3.3403	0.4837 + j3.3316	0.5318 + j4.4539	7.21	7.52	12.22	13.8

J2304 POI Without J2304	0.8410 + j13.6787	0.9489 + j13.7432	4.9021 + j29.1001	16.27	8.45	18.03	9.54
J2304 POI With J2304	0.6661 + j12.1777	0.7508 + j12.2300	2.3960 + j20.4828	18.28	11.77	18.03	17.2
J2316 POI Without J2316	1.6637 + j11.9737	1.7026 + j12.0026	1.5107 + j11.6139	7.2	7.3	8.5	10.08
J2316 POI With J2316	1.6637 + j11.9737	1.7026 + j12.0026	1.5107 + j11.6139	7.2	7.3	8.5	10.08

Table 4.1-4 – Minimum Fault Thevenin Equivalent Data

Location	Positive Sequence (Ohm)	Negative Sequence (Ohm)	Zero Sequence (Ohm)	Three- Phase Theveni n X/R Ratio	Single- Phase Theveni n X/R Ratio	Three- Phase ANSI 37.010 X/R Ratio	Single- Phase ANSI 37.010 X/R Ratio
J1895 POI Without J1895	2.5083 + j17.5911	2.5202 + j17.5918	4.6492 + j30.3909	7.01	6.78	7.19	9.78
J1895 POI With J1895	2.5083 + j17.5911	2.5202 + j17.5918	0.4821 + j9.5175	7.01	8.11	7.19	8.62
J1931 POI Without J1931	3.9389 + j36.0172	3.9431 + j36.0232	1.7261 + j25.3700	9.14	10.14	9.27	10.95
J1931 POI With J1931	3.9389 + j36.0172	3.9431 + j36.0232	1.5643 + j24.1543	9.14	10.18	9.27	10.91
J1934 POI Without J1934	2.3862 + j15.4456	2.4247 + j15.4453	6.6785 + j29.9481	6.47	5.3	6.61	5.14
J1934 POI With J1934	2.3862 + j15.4456	2.4247 + j15.4453	1.1900 + j12.8366	6.47	7.29	6.61	8.51
J1935 POI Without J1935	1.2282 + j4.7986	1.2347 + j4.8126	3.9868 + j14.9018	3.91	3.8	4.18	3.98
J1935 POI With J1935	1.2282 + j4.7990	1.2348 + j4.8130	1.0550 + j7.7464	3.91	4.93	4.18	6.21
J1943 POI Without J1943	1.5550 + j16.8941	1.5848 + j16.8976	1.5227 + j14.8315	10.86	10.43	14.99	14.65
J1943 POI With J1943	1.5550 + j16.8941	1.5848 + j16.8976	1.1876 + j13.0794	10.86	10.83	14.99	14.96
J1993 POI Without J1993	3.0237 + j44.2360	3.0430 + j44.2300	4.6995 + j54.0561	14.63	13.24	16.81	19.2
J1993 POI With J1993	3.0237 + j44.2360	3.0430 + j44.2300	2.9012 + j42.4646	14.63	14.6	16.81	19.46
J2029 POI Without J2029	4.2701 + j23.7216	4.5761 + j23.8973	13.1512 + j53.2167	5.56	4.58	6.65	5.08
J2029 POI With J2029	4.2701 + j23.7216	4.5761 + j23.8973	3.6674 + j27.3172	5.56	5.99	6.65	7.52
J2055 POI Without J2055	1.4180 + j17.9664	1.4694 + j17.9685	3.8032 + j29.7265	12.67	9.81	27.48	23.43
J2055 POI With J2055	1.4180 + j17.9664	1.4694 + j17.9685	3.8031 + j29.7258	12.67	9.81	14.5	16.36
J2099 POI Without J2099	3.0921 + j38.6358	3.1517 + j38.6178	0.9355 + j17.2141	12.5	13.16	12.91	13.21
J2099 POI With J2099	3.0921 + j38.6358	3.1517 + j38.6178	0.8899 + j13.4916	12.5	12.72	12.91	12.84
J2107 POI Without J2107	3.0921 + j38.6358	3.1517 + j38.6178	0.9355 + j17.2141	12.5	13.16	12.91	13.21
J2107 POI With J2107	3.0921 + j38.6358	3.1517 + j38.6178	0.8899 + j13.4916	12.5	12.72	12.91	12.84
J2148 POI Without J2148	0.8827 + j7.0254	0.8860 + j7.0188	3.4842 + j14.7947	7.96	5.49	7.68	6.14
J2148 POI With J2148	0.8827 + j7.0254	0.8860 + j7.0188	1.7130 + j10.1065	7.96	6.94	7.68	8.19
J2185 POI Without J2185	3.0921 + j38.6358	3.1517 + j38.6178	0.9355 + j17.2141	12.5	13.16	12.91	13.21

J2185 POI With J2185	3.0921 + j38.6358	3.1517 + j38.6178	0.8899 + j13.4916	12.5	12.72	12.91	12.84
J2193 POI Without J2193	6.4875 + j24.5323	6.4910 + j24.5356	5.3720 + j32.0617	3.78	4.42	3.97	5.64
J2193 POI With J2193	6.4875 + j24.5323	6.4910 + j24.5356	1.1274 + j14.1921	3.78	4.48	3.97	4.87
J2211 POI Without J2211	0.5885 + j4.0710	0.6077 + j4.0627	1.5709 + j8.6324	6.92	6.06	9.69	8.58
J2211 POI With J2211	0.5885 + j4.0710	0.6077 + j4.0627	0.6015 + j5.3442	6.92	7.5	9.69	11.83
J2304 POI Without J2304	2.0874 + j33.9479	2.1650 + j33.9635	9.4567 + j67.8204	16.26	9.9	17.08	11.31
J2304 POI With J2304	2.0874 + j33.9479	2.1650 + j33.9635	2.3578 + j34.0714	16.26	15.43	17.08	18.81
J2316 POI Without J2316	4.0622 + j26.2121	4.1326 + j26.2695	6.8805 + j48.0498	6.45	6.67	6.91	9.38
J2316 POI With J2316	4.9082 + j29.4961	5.0472 + j29.5898	1.0265 + j16.8367	6.01	6.91	6.91	8.05

ATC breaker duty analysis results for over duty breakers are summarized in Table 4.1-5. Based on the results, no ATC circuit breakers exceeded the over duty Equipment Rating in CAPE.

Table 4.1-5 – Over Duty Breakers Found in SLG Analysis
None

Please note, no breaker duty analysis was performed for any affected system equipment. Appendix N provides short circuit data for all ATC-owned substation buses 69 kV and above included in ATC protection CAPE models. Any non-ATC equipment requiring a change would be considered Affected System.

4.2 Network Upgrades Identified in the Short Circuit Study

Table 4.2-1 Short circuit Network Upgrades Identified

None

5.0 TRANSFORMER ENERGIZATION ANALYSIS

5.1 Transformer Energization Study Results

The Transformer Energization Analysis was performed based on ATC transformer initial energization criteria as described in Appendix A. The 2026 Shoulder Batteries Discharging model was used with all identified ERIS and NRIS Network Upgrades included. The Phase 3 results are summarized in Table 5.1-1. No constraints were found as all inrush voltages were greater than the allowed minimum of 0.70 pu. If during later study phases any project is determined to have an inrush voltage below 0.70 pu the project can go forward if this issue is addressed by (1) a detailed EMT study commissioned by the developer determines the minimum inrush voltage is acceptable, (2) the generator changes their project's design to increase the minimum inrush voltage (increasing the transformer impedance), or (3) controlled closing is used when energizing the transformer to reduce inrush currents to acceptable levels.

Table 5.1-1 – Inrush Calculations Using Shoulder Model

Generation Project	Transformer(s)		PSSE POI	Fault	Vmin Inrush (pu)	
	Number	Windings	Bus #	Current (A)	Raw	Multiplier
J1895	1	3	693710	8338	0.7104	0.8170
J1931	1	3	693863	12070	0.9818	1.1291
J1934	1	3	699362	14562	0.8770	1.0086
J1935	1	3	698179	20192	0.9325	1.0723
J1943	2	3	999907	16369	0.9671	1.1122
J1993	1	3	46294	11901	0.9604	1.1044
J2029	2	3	699894	4164	0.9250	1.0638
J2055	1	3	694010	12589	0.8973	1.0319
J2099	2	3	50997	14321	0.8896	1.0230
J2107	2	3	50997	14321	0.8896	1.0230
J2148	1	3	698929	17919	0.9104	1.0470
J2185	2	3	50997	14321	0.8896	1.0230
J2193	1	3	699939	5235	0.7000	0.8050
J2211	1	3	699611	15537	0.8199	0.9429
J2304	2	3	53047	112118	0.9395	1.0804
J2316	1	3	699566	5840	0.7890	0.9074

6.0 PSCAD ANALYSIS

ATC performed a low short circuit strength screening analysis for all inverter-based resources in the DPP 2021 East (ATC) Phase 1 which determined that the following requests that have proceeded to Phase 3 would require further electromagnetic transient (EMT) analysis in PSCAD:

- J1895
- J1931
- J1943
- J1993
- J2029
- J2099
- J2107
- J2148
- J2185
- J2193
- J2304
- J2316

PSCAD requires much more data and processing time than traditional steady state and dynamics software packages used in transmission planning such that system wide area models do not exist. To study the generators, seven different local models were produced and listed below.

- Southwest Wisconsin (SW WI) 138 kV
- Southwest Wisconsin (SW WI) 345 kV
- Port Edwards – Wautoma (POE WAU) 138 kV
- Southcentral Wisconsin (SC WI) 138 kV
- Northwest Upper Peninsula (NW UP) 138 kV
- ATC Planning Zone 1 (Z1) 345 kV

ATC has determined through numerous studies that inverter-based resources can be sensitive to local cap bank dispatch in PSCAD studies. Based on engineer judgement, ATC selects a conservative cap bank dispatch scenario for each PSCAD model. Table 6.0-1 details the local cap banks dispatched in each of the PSCAD models.

Table 6.0-1 – Cap Bank Dispatches for DPP-2021 PSCAD Cases

Model	IBR Plants Included	Cap Banks to Dispatch
SW Wisconsin 138 kV	Quilt Block Wind, Eden Wind, J855, J864, J870, J871, J947, J1000, J1188, J1304, J1305, J1374, J1483, J1706, J1773, J1781, J1931	J1773 34.5 kV 2 x 10 MVARs J1483 34.5 kV 1 x 3 MVARs J1781 34.5 kV 2 x 10 MVARs Kirkwood 138 kV 1 x 19.2 MVARs Spring Green 138 kV 1 x 18 MVARs Eden 138 kV 1 x 19.2 MVARs North Monroe 138 kV 26.4 MVARs Badger Hollow 34.5 kV 4 x 12.6 MVARs Red Barn 34.5 kV 1 x 5 MVARs J1374 34.5 kV 2 x 2 MVARs Grant County Solar 34.5 kV 2 x 14 MVARs

		Quiltblock Wind 4 x 11 MVARs (only dispatch to stay in rated Q limits)
SW Wisconsin 345 kV	J1214, J1326, J1377, J1410, J1411, J1483, J1629, J1746, J1773, J1779, J1781, J1931, J1993, J2304	J1773 34.5 kV 2 x 10 MVARs J1483 34.5 kV 1 x 3 MVARs J1781 34.5 kV 2 x 10 MVARs J1214 34.5 kV 2 x 4 MVARs J1779 34.5 kV 2 x 4 MVARs J1410 34.5 kV 2 x 4 MVARs J1411 34.5 kV 1 x 4 MVARs J1993 34.5 kV 1 x 30 MVARs
Port Edwards - Wautoma 138 kV	J986, J1002, J1751, J2117, J2193	J1751 34.5 kV 1 x 29 MVARs Port Edwards 138 kV 1 x 18 MVARs Wautoma 138 kV 2 x 24.5 MVARs Wautoma 69 kV 1 x 16.3 MVARs McKenna 69 kV 1 x 15.3 MVARs Chaffee Creek 69 kV 1 x 14.4 MVARs Oxford 69 kV 1 x 6.3 MVARs Wautoma Solar 34.5 kV 1 x 8 MVARs Wood County Solar 34.5 kV 1 x 19.9 MVARs
South Central Wisconsin 138 kV	J850, J1895	State Line 138 kV 2 x 16.3 MVARs (693421) N Lake Geneva 69 kV 1 x 10.8 MVARs Balsam 138 kV 2 x 16.3 MVARs
NW UP 138 kV	J1814, J2029, J2316	Plains 138 kV 3 x 19.8 MVARs National 138 kV 2 x 20.4 MVARs
Zone 1 345 kV	J1513, J1567, J2099, J2107, J2148, J2185	none

6.1 SW WI 138 kV PSCAD Study

Full details of the SW WI 138 kV PSCAD study can be found in Appendix L.

6.1.1 SW WI 138 kV PSCAD Study Results

Two P6 faults were found to be unstable. The P6.1.1 fault of a [REDACTED] [REDACTED] was found to be unstable. The P6.1.1 fault of a [REDACTED] [REDACTED] followed by a fault on Hill Valley – Hickory Creek 345 kV was found to be unstable. All other faults studied were found to be stable. The results are summarized in Table 6.1.1-1. Plots of the SW WI 138 kV analysis can be found in Appendix L.

Table 6.1.1-1 – SW WI 138 kV PSCAD Results

TPL	Fault Description	Fault Type	Plants Monitored ⁰
P1.2		3PG	OK
		1PG	OK
P1.2		3PG	OK
		1PG	OK
P1.2		3PG	OK
		1PG	OK
P6.1.1		3PG	UNSTABLE ¹
		1PG	UNSTABLE ¹
P6.1.1		3PG	UNSTABLE ²
		1PG	UNSTABLE ²

⁰ Redbarn Wind (J855), Badger Hollow Solar (J870/J871), Grant County Solar (J947), Bear Creek Solar (J864), North Rock Solar (J1188), Quiltblock Wind Farm (J395), Montfort Wind Farm, Cassville Solar (J1000), Albany Solar (J1374), Paddock Solar (J1305), J1483, J1706, J1773, J1781, J1931, and Riverside CTs 1 & 2 were monitored.

¹ All plants experienced oscillations greater than 3% and or tripped

² Nelson Dewey - Eden – Hill Valley area and Badger Hollow plants: depressed voltage (~0.85 pu) for 10s. Some oscillations present during undervoltage. Quiltblock Wind power recovery takes longer than 1s. Several plants stay in an LVRT mode for several seconds.

6.1.2 Network Upgrades Identified in the SW WI 138 kV PSCAD Study

No network upgrades were identified to mitigate constraints from the SW WI PSCAD study.

6.1.3 Operating Restrictions Identified in the SW WI 138 kV PSCAD Study

The two P6 contingencies studied could require operating restrictions of the unstable plants down to as low as 0 MW to mitigate the instability after a prior outage. Given the increasing complexity of the SW WI PSCAD study and the expectation that plant models in this area will be changing with generator modification submittals and as-built parameters, these operating restrictions were not calculated as part of this DPP study and will be performed at a later stage of the interconnection process.

6.2 SW WI 345 kV PSCAD Study

Full details of the SW WI 345 kV PSCAD study can be found in Appendix L.

6.2.1 SW WI 345 kV PSCAD Study Results

All faults studied in the Southwest Wisconsin 345 kV area were found to be stable. The results are summarized in Table 6.2.1-1. Plots of the SW WI 345 kV analysis can be found in Appendix L.

Table 6.2.1-1 – SW WI 345 kV PSCAD Results

[illegible]

		1PG	OK
		3PG	OK
P6.1.1		1PG	OK
		3PG	OK
P6.1.1		1PG	OK

⁰ J1214, J1326, J1377, J1410, J1411, J1483, J1502, J1629, J1746, J1773, J1779, J1781, J1931

J1943, J1993, J2304, Byron, and Riverside CTs 1 & 2 were monitored.

6.2.2 Network Upgrades Identified in the SW WI 345 kV PSCAD Study

No network upgrades were identified to mitigate constraints from the SW WI 345 kV PSCAD study.

6.2.3 Operating Restrictions Identified in the SW WI 345 kV PSCAD Study

No operating restrictions were identified to mitigate constraints from the SW WI 345 kV PSCAD study.

6.3 POE WAU 138 kV PSCAD Study

Full details of the Port Edwards – Wautoma 138 kV PSCAD study can be found in Appendix L.

6.3.1 POE WAU 138 kV PSCAD Study Results

J2117 withdrew from DPP-2021 phase 3. ATC assessed the impact from this withdrawal and given the fact that the request was a 0 MW ERIS/NRIS request ATC did not find that the withdrawal would make the IBR stability in the area worse. Consequently, the Port Edwards – Wautoma area PSCAD analysis remains unchanged from the analysis presented in DPP-2021 phase 2.

Two P1 faults and nine P6 faults were found to be unstable. All other faults studied were found to be stable. Plots of the POE WAU 138 kV analysis can be found in Appendix L.

Since two of the unstable results were for P1 events ATC cannot mitigate this instability by operating restrictions. Therefore, the P1 events were rerun with a typical cap bank dispatch instead of a worst case cap bank dispatch. All of the P1 faults were found to be stable in the typical cap bank dispatch scenario. Those results are summarized in Table 6.3.1-2.

A sensitivity analysis was also performed for the J1751/J2117 with a new Colburn T-D project added to the system. The Colburn T-D will network Line Y-302 into the Badger West substation. Two of the P6 faults studied were unstable. Two P6 faults were found to be unstable. All other faults studied were found to be stable. The results are summarized in Table 6.3.1-3. Plots of the SC WI 138 kV analysis can be found in Appendix L.

Table 6.3.1-1 – POE WAU 138 kV PSCAD Results with Worst Case Cap Bank Dispatch

TPL	Fault Description	Fault Type	Plants Monitored ⁰
P1.2		3PG	UNSTABLE ¹
		1PG	UNSTABLE ²
P1.2		3PG	OK
		1PG	OK
P1.2		3PG	OK
		1PG	OK
P1.2		3PG	OK
		1PG	OK
P6.2.1		3PG	UNSTABLE ⁴
		1PG	UNSTABLE ⁴
P6.1.1		3PG	UNSTABLE ⁵
		1PG	UNSTABLE ⁵
P6.2.1		3PG	UNSTABLE ⁴
		1PG	UNSTABLE ⁴
P6.1.1		3PG	UNSTABLE ⁵
		1PG	UNSTABLE ⁵
P6.2.1		3PG	OK
		1PG	OK
P6.2.1		3PG	OK
		1PG	OK
P6.1.1		3PG	UNSTABLE ⁶
		1PG	UNSTABLE ⁶
P6.2.1		3PG	OK
		1PG	OK
P6.1.1		3PG	UNSTABLE ⁵
		1PG	UNSTABLE ⁵
P6.2.1		3PG	OK
		1PG	OK
P6.1.1		3PG	UNSTABLE ⁶
		1PG	UNSTABLE ⁶
P6.1.1		3PG	UNSTABLE ⁷
		1PG	UNSTABLE ⁷
P1.2		3PG	UNSTABLE ³
		1PG	UNSTABLE ³
P6.2.1		3PG	UNSTABLE ⁸
		1PG	UNSTABLE ⁸
P1.2		3PG	OK
		1PG	OK
P6.1.1		3PG	OK

		1PG	OK
		3PG	OK
P6.1.1		1PG	OK

⁰ Wood County Solar (J986), Wood County BESS (S1018), Wautoma Solar (J1000), Wautoma BESS (S1024), J2193, J1751, J2117, and J2293.

¹ After fault clearing, J986, J1002, and J2193 all have large oscillations for the duration of the simulation.

² All plants initially recover after the fault. J986, J1002, and J2193 start oscillating at 11s for about 4 seconds before again recovering by 20s.

³ J1751 / J2117 do not recover after the fault and oscillate unstably for remainder of simulation.

⁴ After fault clearing, J986, J1002, J2193 and J2293 all have large oscillations for the duration of the simulation.

⁵ J986 trips at fault clearing. J2193, J1002, and J2293 all have large oscillations for the duration of the simulation.

⁶ J1002, J2193, and J2293 all have large oscillations for the duration of the simulation

⁷ J986 trips at fault clearing. J1002 trips 3.5s after fault due to overvoltage. J2293 trips 10s after fault due to overvoltage.

⁸ J1751 / J2117 trips and then attempts to recover but is unstable several times over the 20s post-fault window. J2293 also has some oscillations / instability.

Table 6.3.1-2 – POE WAU 138 kV PSCAD Results with Typical Cap Bank Dispatch

TPL	Fault Description	Fault Type	Plants Monitored ⁰
P1.2		3PG	OK
		1PG	OK
P1.2		3PG	OK
		1PG	OK
P1.2		3PG	OK
		1PG	OK
P1.2		3PG	OK
		1PG	OK
P1.2		3PG	OK
		1PG	OK
P1.2		3PG	OK
		1PG	OK

⁰ Wood County Solar (J986), Wood County BESS (S1018), Wautoma Solar (J1000), Wautoma BESS (S1024), J2193, J1751, J2117, and J2293.

Table 6.3.1-3 – POE WAU 138 kV PSCAD Results with Colburn T-D and Worst Case Cap Bank Dispatch

TPL	Fault Description	Fault Type	Plants Monitored ⁰
P1.2		3PG	OK
		1PG	OK
P6.2.1		3PG	UNSTABLE ⁸
		1PG	UNSTABLE ⁸

P6.1.1		3PG	UNSTABLE ⁹
		1PG	UNSTABLE ⁹

⁰ Wood County Solar (J986), Wood County BESS (S1018), Wautoma Solar (J1000), Wautoma BESS (S1024), J2193, J1751, J2117, and J2293.

⁸ J1751 / J2117 trips and then attempts to recover but is unstable several times over the 20s post-fault window. J2293 also has some oscillations / instability.

⁹ J1751 / J2117 trips due to overvoltage.

6.3.2 Network Upgrades Identified in the POE WAU 138 kV PSCAD Study

Since all of the P1 events were stable in the typical cap bank dispatch scenario, no network upgrades were identified to mitigate constraints from the POE WAU 138 kV PSCAD study.

6.3.3 Operating Restrictions Identified in the POE WAU 138 kV PSCAD Study

The unstable P6 contingencies studied could require operating restrictions of the unstable plants down to as low as 0 MW to mitigate the instability after a prior outage. Given the expectation that plant models in this area will be changing with generator modification submittals and as-built parameters, these operating restrictions were not calculated as part of this DPP study and will be performed at a later stage of the interconnection process.

6.4 SC WI 138 kV PSCAD Study

Full details of the Southcentral Wisconsin 138 kV PSCAD study can be found in Appendix L.

6.4.1 SC WI 138 kV PSCAD Study Results

Two P6 faults were found to be unstable. All other faults studied were found to be stable. The results are summarized in Table 6.4.1-1. Plots of the SC WI 138 kV analysis can be found in Appendix L.

Table 6.4.1-1 – SC WI 138 kV PSCAD Results

TPL	Fault Description	Fault Type	Plants Monitored ⁰
P1.2		3PG	OK
P1.2		1PG	OK
		3PG	OK
P1.2		1PG	OK
		1PG	OK
P4.2		1PG	OK
P4.3		1PG	OK
P4.2		1PG	OK
P4.3		1PG	OK
P6.1.1		3PG	OK
	1PG	OK	

P6.1.1		3PG	OK
		1PG	OK
P6.1.1		3PG	UNSTABLE ¹
		1PG	UNSTABLE ¹
P6.1.1		3PG	UNSTABLE ¹
		1PG	UNSTABLE ¹
P6.1.1		3PG	OK
		1PG	OK
P6.1.1		3PG	OK
		1PG	OK

⁰ Darien Solar (J850) and J1895.

¹ A large overvoltage with distortions observed at J1895 POI after fault clearing, which cause J1895 to trip shortly after (occurring due to 69kV path being the only remaining outlet)

6.4.2 Network Upgrades Identified in the SC WI 138 kV PSCAD Study

No network upgrades were identified to mitigate constraints from the SC WI 138 kV PSCAD study.

6.4.3 Operating Restrictions Identified in the SC WI 138 kV PSCAD Study

The unstable P6 contingencies studied could require operating restrictions of the unstable plants down to as low as 0 MW to mitigate the instability after a prior outage. Given the expectation that plant models in this area will be changing with generator modification submittals and as-built parameters, these operating restrictions were not calculated as part of this DPP study and will be performed at a later stage of the interconnection process.

6.5 NW UP 138 kV PSCAD Study

Full details of the Northwest Upper Peninsula 138 kV PSCAD study can be found in Appendix L.

6.5.1 NW UP 138 kV PSCAD Study Results

All faults studied in the Northwest Upper Peninsula 138 kV area were found to be stable. The results are summarized in Table 6.5.1-1. Plots of the NW UP 138 kV analysis can be found in Appendix L.

Table 6.5.1-1 – NW UP 138 kV PSCAD Results

TPL	Fault Description	Fault Type	Plants Monitored ⁰
P6.1.1		3PG	OK
		1PG	OK
P6.1.1		3PG	OK
		1PG	OK

⁰ 1750, J2029, J2316, and National STATCOM were monitored

6.5.2 Network Upgrades Identified in the NW UP 138 kV PSCAD Study

No network upgrades were identified to mitigate constraints from the NW UP 138 kV PSCAD study.

6.5.3 Operating Restrictions Identified in the NW UP 138 kV PSCAD Study

No operating restrictions were identified to mitigate constraints from the NW UP 138 kV PSCAD study.

6.6 Z1 345 kV PSCAD Study

Full details of the ATC Planning Zone 1 345 kV PSCAD study can be found in Appendix L.

6.6.1 Z1 345 kV PSCAD Study Results

One P6 fault was found to be unstable. All other faults studied were found to be stable. The results are summarized in Table 6.6.1-1. Plots of the Z1 345 kV analysis can be found in Appendix L.

Table 6.6.1-1 – Z1 345 kV PSCAD Results

TPL	Fault Description	Fault Type	Plants Monitored ⁰
P1.2		3PG	OK
		1PG	OK
P1.2		3PG	OK
		1PG	OK
P1.3		3PG	OK
		1PG	OK
P6.1.1		3PG	UNSTABLE ¹
		1PG	UNSTABLE ²
P6.1.1		3PG	OK
		1PG	OK
P6.1.1		3PG	OK
		1PG	OK
P6.1.1		3PG	OK
		1PG	OK
P6.1.1		3PG	OK
		1PG	OK
P6.1.2		3PG	OK
		1PG	OK

⁰ J1513, J2148, J2099, J2107, and J2185 were monitored.

¹ Following fault clearing severe oscillations in active and reactive power are present at all the monitored IBR POIs for approximately 7s after fault clearing

² Following fault clearing severe undamped oscillations in active and reactive power are present at all the monitored IBR terminals

6.6.2 Network Upgrades Identified in the Z1 345 kV PSCAD Study

No network upgrades were identified to mitigate constraints from the Z1 345 kV PSCAD study.

6.6.3 Operating Restrictions Identified in the Z1 345 kV PSCAD Study

The unstable P6 contingency studied could require operating restrictions of the unstable plants down to as low as 0 MW to mitigate the instability after a prior outage. Given the expectation that plant models in this area will be changing with generator modification submittals and as-built parameters, these operating restrictions were not calculated as part of this DPP study and will be performed at a later stage of the interconnection process.

7.0 AFFECTED SYSTEM ANALYSIS

7.1 PJM Affected System Steady State Study Summary

PJM performed the PJM Affected System analysis, and the report can be found in Appendix H. No PJM Affected System Network Upgrades have cost allocations to the study generators in the DPP-2021 Phase 3 East (ATC) group.

7.2 AECI Affected System Steady State Study Summary

Associated Electric Cooperative Inc. (AECI) performed the AECI Affected System analysis, and the report can be found in Appendix H. No AECI Affected System Network Upgrades have cost allocations to the study generators in the DPP-2021 Phase 3 East (ATC) group.

7.3 WPS Affected System Steady State Study Summary

Based on the Phase 3 short circuit study results, Wisconsin Public Service (WPS) provided the following estimated upgrade scopes and associated cost estimates for mitigating increased fault current at the following substations. WPS Affected System Upgrades have cost allocations to J2055, J2099, J2107, J2148, J2185, and J2211. A summary of cost allocations is provided in Section 10 Table 10.4-3.

Table 7.3-1 – WPS Affected System Upgrades

Substation	Ownership Type	Bus	kV	Total Estimated Cost (\$)	Scope of Work
PACKAGING	JUCR*	341500	138	124,000	Ground Grid Mitigation
PULLIAM	JUCR*	387800	138	230,000	Ground Grid Mitigation
HIGHWAY V	JUCR*	961001	138	124,000	Ground Grid Mitigation
ONTARIO	JUCR*	961400	138	127,000	Ground Grid Mitigation
VELP AVE	JUCR*	960800	138	124,000	Ground Grid Mitigation
JAMES ST.	JUCR*	960100	138	127,000	Ground Grid Mitigation

EASTMAN AVE.	JUCR*	961300	138	124,000	Ground Grid Mitigation
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*JUCR - Joint Use Common Retained

7.4 We Energies Affected System Steady State Study Summary

Based on the Phase 3 short circuit study results, We Energies provided the following estimated upgrade scopes and associated cost estimates for mitigating increased fault current at the following substations. WE Affected System Upgrades have cost allocations to J1895, J1934, J1943, J1993, J2055, J2099, J2107, J2148, J2185, J2211, and J2304. A summary of cost allocations is provided in Section 10 Table 10.4-4.

Table 7.4-1 – We Energies Affected System Upgrades

Substation	Ownership Type	Bus	kV	Total Estimated Cost (\$)	Scope of Work
TAMARACK	JUCR*	950800	138	488,000	Circuit Switcher Replacement, Ground Grid Mitigation
GERMANTOWN	JUCR*	931900	138	138,000	Ground Grid Mitigation
HINTZ	JUCR*	953400	138	138,000	Ground Grid Mitigation
SPRING VALLEY WE	JUCR*	851800	138	138,000	Ground Grid Mitigation
MAPLE	JUCR*	938000	138	138,000	Ground Grid Mitigation
BUTLER	JUCR*	926800	138	138,000	Ground Grid Mitigation

*JUCR - Joint Use Common Retained

8.0 DELIVERABILITY STUDY

Generator interconnection requests have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS). If the generator is determined as not fully deliverable, the customer can either choose to elect the amount of NRIS available without upgrades or build system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up.

MISO Generator Deliverability Study methodology is described in MISO BPM-15.

8.1 Study Summary

The summary of MISO deliverability results based on the 2026 summer peak study model is shown in the following tables. Detailed NRIS study results and individual generator summaries can be found in Appendix I.

Table 8.1-1 below lists the deliverability results with ERS upgrades included in the NRIS analysis. Deliverable NRIS is the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction (Conditional on ERS upgrades only and case assumptions).

Table 8.1-1 – NRIS Analysis Summary

MISO Queue #	Area	Requested NRIS (MW)	Deliverable NRIS Amount in SUM 2026 Case (MW)*
J1895	ATC	200	61.25
J1931	ATC	12	3.68
J1934	ATC	150	45.94
J1935	ATC	50	15.31
J1943	ATC	200	61.26
J1993	ATC	99	30.32
J2029	ATC	62.5	19.14
J2055	ATC	100	30.63
J2099	ATC	195.79	59.97
J2107	ATC	195.79	59.97
J2148	ATC	150	45.94
J2185	ATC	195.79	59.97
J2193	ATC	100	30.63
J2211	ATC	200	61.25
J2304	ATC	400	122.5
J2316	ATC	71.5	21.9

*This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction (Conditional on ERS upgrades only and case assumptions)

Table 8.1-2 below lists all the NRIS constraints from the deliverability study and the identified NRIS Network Upgrades. Both ERS Network Upgrades and NRIS Network Upgrades must be made for 100% NRIS, i.e. fully deliverable. Please note, if a NRIS Network Upgrade entirely or partially changes the scope of an ERS Network Upgrade, only the cost difference between the NRIS upgrade and the ERS upgrade will be eligible for NRIS Network Upgrade cost allocation.

Due to the queue size and a large number of study generators in different MISO regions that were identified to be responsible for the NRIS Network Upgrades, considering readability, both the individual generator deliverability report and the detailed NRIS Network Upgrade cost allocation information including worst MW impact and cost allocation percentage are provided in Appendix I.

Table 8.1-2 – Network Upgrades Needed to Address MISO Identified NRIS Steady-State Injection Constraints

NRIS Thermal Constraints	Required NRIS Network Upgrades	Network Upgrade Owner	NRIS Network Upgrade Cost Estimates (\$) [a]	ERIS Network Upgrade (Identified for Same Constraint) Cost Estimate (\$) [b]	Previous Cycle ERIS/NRIS Network Upgrade (Identified for Same Constraint) Cost Estimate (\$) [c]	Cost Used for NRIS Network Upgrade Allocation (\$) [d] = [a] - [b] - [c]
699167 COL 138 138 699142 PORTAGE B2 138 2	MTEP PID 50673 Portage - Columbia 138 kV (X-13/X-20), Reconductor	ATC	0	0	0	0
699167 COL 138 138 699145 POR 138 138 1						
694119 RACINE BUS5 345 693481 MT PLEASANT 345 1	Elm Rd – Jupiter 345 kV, New Line	ATC	234,967,148	0	0	234,967,148
694121 RACINE BUS7 345 693481 MT PLEASANT 345 2						
699367 ELM ROAD 345 694117 RACINE BUS1 345 1						
699367 ELM ROAD 345 694117 RACINE BUS1 345 2						
695053 MILL RD B2 345 699247 ARCADIAN 345 345 1	MTEP PID 50562 LRTP 29: Granville – Bluemound 345 kV Line ¹	ATC	0	0	0	0
695058 MILL RD B7 345 699247 ARCADIAN 345 345 2						
699432 PLS PR1 345 270941 ZION STA ; R 345 1	MTEP PID 50562 LRTP 29: Lakeview 345 kV Substation	ATC	0	0	0	0
699432 PLS PR1 345 274817 ZION EC ;RP 345 1						
695110 LAKEVIEW 345 270942 ZION STA ;OB 345 1	Lakeview – Zion 345 kV (2222), Uprate ²	ComEd	2,200,000	0	0	2,200,000
43388 J1363 POI 345 43524 J1352 POI 345 1	Montgomery to Spencer Creek rebuild (J1363 POI - J1352 POI) ³	AMMO	5,400,000	0	0	5,400,000
345992 7SPENCER 345 43386 J1338 POI 345 1	Montgomery to Spencer Creek rebuild (Spencer - J1338 POI) ³	AMMO	5,400,000	0	0	5,400,000
43386 J1338 POI 345 43388 J1363 POI 345 1	Montgomery to Spencer Creek rebuild (J1338 POI - J1363 POI) ³	AMMO	32,400,000	0	0	32,400,000
255113 17STILLWELL 345 243219 05DUMONT 345 1	DPP20 Central NU: Rebuild Stillwell - Dumont	NIPS	0	0	0	0

¹ATC proposed Granville – Bluemound 345 kV Line as an NRIS Network Upgrade. Per MISO BPM 15, the LRTP Tranche 2.1 Project 29 (MTEP PID 50562) contains the same facility as the identified network upgrade and has therefore been cited.

²Mitigation is for ComEd scope only. The ATC end of the line requires no work and will be a rating correction. ComEd scope is sag mitigation of ~2 miles of conductor (~8 possible impacted towers) and a line trap replacement for \$2,200,000.

³These NUs are listed here as placeholders only. Please refer to the DPP-2021-Phase3-Central SIS report when available for potential revisions.

8.2 Network Upgrade Alternatives Considered

All NRIS network upgrades identified in Table 8.1-2, except for “Elm Rd – Jupiter 345 kV, New Line” are direct upgrades of the constraint facilities to ATC design standards and considered as least-cost solutions. Therefore, no other alternatives were examined for those solutions.

For “Elm Rd – Jupiter 345 kV, New Line” one alternative was considered. The alternative was to build a new substation at the junction of the existing Elm Rd – Arcadian 345 kV (L-ERG71) line and existing Elm Road – Racine 345 kV (L-ERG91) line. A new 345 kV line is then built from the new substation to the Jupiter substation. This option performed similarly to the “Elm Road – Jupiter 345 kV, New Line” option, but is more costly. Therefore, ATC decided on “Elm Rd – Jupiter 345 kV, New Line” as the required network upgrade.

9.0 SHARED NETWORK UPGRADE ANALYSIS

Shared Network Upgrade (SNU) Analysis, which tests for Network Upgrades driven by higher queued interconnection projects, was performed for this System Impact Study. No generators were found to meet the requirements to share costs of Network Upgrades driven by higher queued interconnection projects.

10.0 COST ALLOCATION AND TRANSMISSION OWNER SELF-FUND ELECTION

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the System Impact Study report date. The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

Per MISO tariff, Transmission Owners must provide a non-binding indication of whether or not they will self-fund any Network Upgrades identified in the System Impact Study analysis. The Self-Funding elections are shown in sections 10.1, 10.2, and 10.3.

10.1 Interconnection Facilities Proposed for DPP 2021 Cycle Phase 3 Projects

For DPP 2021 Cycle Phase 3 East (ATC) group, the total costs of interconnection facilities are summarized in Table 10.1-1.

Table 10.1-1 – Interconnection Facilities for DPP 2021 Cycle Phase 3 Projects

MISO Queue #	Interconnection Facilities (\$)	
	TO Network Upgrades	Self-Fund
J1895	4,923,000	Yes
J1931	25,000	Yes
J1934	5,755,000	Yes
J1935	940,000	Yes
J1943	225,000	Yes
J1993	225,000	Yes
J2029	2,602,000	Yes
J2055	4,538,000	Yes
J2099	9,164,000	Yes
J2107	9,164,000	Yes
J2148	1,618,000	Yes
J2185	9,164,000	Yes
J2193	8,862,000	Yes
J2211	1,304,000	Yes
J2304	31,578,900	Yes
J2316	25,000	Yes

10.2 ERIIS Network Upgrades Proposed for DPP 2021 Cycle Phase 3 Projects

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the ERIS analysis. The ERIS network upgrades include thermal network upgrades and voltage support network upgrades identified in the steady-state analysis. For DPP 2021 Cycle Phase 3 East (ATC) group, the total costs of ERIS network upgrades for the 2026 scenario are summarized in Tables 10.2-1, 10.2-2, 10.2-3, 10.2-4, and 10.2-5.

Table 10.2-1 – ERIS Network Upgrades Identified in Steady State Analysis

Network Upgrade	Facility Owner	Cost Used for Cost Allocation (\$) ^{1, 2}	Self-Fund
Pulliam – Tower Dr (V-100) 138 kV, Rebuild	ATC	3,837,891	Yes
Pulliam – Packaging (S-123) 138 kV, Rebuild	ATC	3,849,069	Yes
Tower Dr – Preble (X-154) 138 kV, Rebuild	ATC	10,378,000	Yes
Preble – Highway V (X-158) 138 kV, Rebuild	ATC	8,641,000	Yes
North Appleton – Lost Dauphin (I-113) 138 kV, Rebuild	ATC	25,944,000	Yes
Badger – Cloverleaf - Maplewood (W-75) 138 kV, Uprate	ATC	8,246,546	Yes

Rock River – Townline Road 138 kV, New Line	ATC	9,084,939	Yes
North Lake Geneva - Walworth 69 kV (Y-41), Uprate	ATC	5,581,481	Yes
MTEP PID# 50560 (LRTP-27: Rocky Run - Werner - North Appleton)	ATC	0	N/A
MTEP PID# 23372 (LRTP-05: Tremval – Eau Claire – Jump River)	XEL/ATC	0	N/A
MTEP PID# 20201 (Small Capital Project and Asset Renewal Program 2024)	ATC	0	N/A

¹ ATC Network Upgrades included a contingency based on a project risk generally between 5% – 20%.

² All Network Upgrades were estimated on an estimated ISD assuming GIA execution by 12/31/2025

Table 10.2-2 – EGIS Network Upgrades in Dynamic Stability Analysis
None

Table 10.2-3 – EGIS Network Upgrades Identified in PSCAD Analysis
None

Table 10.2-4 – EGIS Network Upgrades in Short Circuit Analysis
None

Table 10.2-5 EGIS Shared Network Upgrades
None

10.3 NRIS Network Upgrades Proposed for DPP 2021 Cycle Phase 3 Projects

Network upgrades for Network Resource Interconnection Service (NRIS) were identified in the MISO's deliverability analysis and listed in Table 10.3-1 below.

Table 10.3-1 – NRIS Network Upgrades Identified

Network Upgrade	Network Upgrade Owner	Cost Used for Cost Allocation (\$) ^{1, 2}	Self-Fund
Elm Rd – Jupiter 345 kV, New Line	ATC	234,967,148	Yes
Lakeview – Zion 345 kV (2222), Uprate	COMED	2,200,000	No ³
MTEP PID 50562 LRTP 29: Granville - Bluemound 345 kV Line	ATC	0	N/A
MTEP PID 50562 LRTP 29: Lakeview 345 kV Substation	ATC	0	N/A
MTEP PID 50673 Portage - Columbia 138 kV (X-13/X-20), Reconductor	ATC	0	N/A
Montgomery to Spencer Creek rebuild (J1363 POI - J1352 POI)	AMMO	5,400,000	Yes
Montgomery to Spencer Creek rebuild (Spencer - J1338 POI)	AMMO	5,400,000	Yes
Montgomery to Spencer Creek rebuild (J1338 POI - J1363 POI)	AMMO	32,400,000	Yes
DPP20 Central NU: Rebuild Stillwell - Dumont	NIPS	0	N/A

¹ ATC Network Upgrades included a contingency based on a project risk generally between 5% – 20%.

² All ATC Network Upgrades were estimated on an estimated ISD assuming GIA execution by 12/31/2025

³ ATC has no scope for the network upgrade; ComEd will not self-fund their portion of the network upgrade.

10.4 Affected System Upgrades Proposed for DPP 2021 Cycle Phase 3 Projects

Table 10.4-1 – PJM Affected System Network Upgrades
None

Table 10.4-2 – AECI Affected System Network Upgrades
None

Table 10.4-3 – WPS Affected System Network Upgrades

Substation	Total Estimated Upgrade Cost (\$)
PACKAGING	124,000
PULLIAM	230,000
VELP AVE	124,000
HIGHWAY V	124,000
ONTARIO	127,000
JAMES ST	127,000
EASTMAN AVE	124,000

Table 10.4-4 – We Energies Affected System Network Upgrades

Substation	Total Estimated Upgrade Cost (\$)
TAMARACK	488,000
GERMANTOWN	138,000
HINTZ	138,000
SPRING VALLEY	138,000
MAPLE	138,000
BUTLER	138,000

10.5 Cost Allocation Methodology for Thermal Network Upgrades

The costs of Network Upgrades (NU) for a set of generation projects (one or more subgroups or entire group with identified NU) are based off the MW impact of the worst-case scenario for each specific generator project. Basically, whatever the highest MW impact (increasing flow) is for that particular generator where the constraint is identified and requires NU is how it should be calculated.

Constraints which are mitigated by one or a subset of NU are identified. The highest MW contribution on these constraints from each generating facility is calculated in the MISO DPP study models without any Network Upgrades. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each generating facility on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contribution on constraint)}} \right)$$

10.6 Cost Estimating and Allocation Methodology for Short Circuit Upgrades

For each ATC breaker shown to be overdutied a new breaker will be scoped and the cost of that upgrade will be assigned to generators. No breaker duty analysis is performed for Affected System equipment. Appendix N provides short circuit data for all substation buses 69 kV and above included in ATC protection CAPE models. Any non-ATC equipment requiring a change would be considered Affected System.

Once ATC breaker replacement costs are determined, they are allocated proportionally to study generators that have a greater than 3% of the total of all current queue study generator contributing fault currents under the single line-ground short circuit fault simulation at the identified substation.

The system impact study does not include any substation ground grid screening study. ATC conducts a ground grid evaluation for ATC brownfield substations in MISO DPP Facility Studies based on available ground grid design documents or in-house models when major substation work is identified. Major substation work includes, but is not limited to:

- Generation addition (brownfield POIs)
- Substation expansion
- Transmission line addition
- Transformer addition or replacement
- Protective device replacement for SLG short-circuit duty reasons

10.7 Cost Allocation Tables

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the issued date of the System Impact Study report.

Due to the queue size and a large number of study generators that were identified to be responsible for the ERIS and NRIS Network Upgrades, considering readability, the detailed ERIS and NRIS Network Upgrade cost allocation information including worst MW impact and cost allocation percentage are provided in Appendices E and I.

Cost allocation tables for ERIS stability Network Upgrades, EMT stability Network Upgrades, short-circuit Network Upgrades, shared network upgrades, and affected system Network Upgrades can be found in tables 10.7.1-8.

Table 10.7-1 – ERIS Stability Network Upgrade Costs Allocated to Each Generation Project

None

Table 10.7-2 – ERIS EMT Stability Network Upgrade Costs Allocated to Each Generation Project

None

Table 10.7-3 – ERIS Short-Circuit Network Upgrade Costs Allocated to Each Generation Project

None

Table 10.7-4 – ERIS Shared Network Upgrade Costs Allocated to Each Generation Project

None

Table 10.7-5 – PJM Affected System Network Upgrades

None

Table 10.7.6 – AECI Affected System Network Upgrades

None

Table 10.7-7 – WPS Affected System Costs Allocated to Each Generation Project

SUBSTATION	Cost Estimate (\$)	Cost Allocation (%)					
		J2055	J2099	J2107	J2148	J2185	J2211
HIGHWAY V	124,000	3.5	4.97	4.97	5.62	4.97	75.97
ONTARIO	127,000	3.62	5.11	5.11	5.75	5.11	75.3
VELP AVE	124,000	-	4.97	4.97	5.78	4.97	79.31
JAMES ST	127,000	-	4.83	4.83	5.53	4.83	79.98
PACKAGING	124,000	-	4.41	4.41	5.03	4.41	81.74
PULLIAM	230,000	-	5.02	5.02	5.74	5.02	79.2
EASTMAN AVE	124,000	-	4.19	4.19	4.78	4.19	82.65

SUBSTATION	Cost Estimate (\$)	Cost Allocation (\$)					
		J2055	J2099	J2107	J2148	J2185	J2211
HIGHWAY V	124,000	4,345	6,166	6,166	6,966	6,166	94,191
ONTARIO	127,000	4,600	6,488	6,488	7,304	6,488	95,632
VELP AVE	124,000	-	6,158	6,158	7,165	6,158	98,361
JAMES ST	127,000	-	6,132	6,132	7,023	6,132	101,581
PACKAGING	124,000	-	5,467	5,467	6,238	5,467	101,361
PULLIAM	230,000	-	11,557	11,557	13,210	11,557	182,119
EASTMAN AVE	124,000	-	5,202	5,202	5,923	5,202	102,471
Total WPS Affected System Upgrade Cost (\$) Allocated to Each		8,945	47,170	47,170	53,829	47,170	775,716

Table 10.7-8 – We Energies Affected System Costs Allocated to Each Generation Project

SUBSTATION	Cost Estimate (\$)	Cost Allocation (%)										
		J1895	J1934	J1943	J1993	J2055	J2099	J2107	J2148	J2185	J2211	J2304
TAMARACK	488,000	10.8	14.34	3.22	3.78	35.83	5.21	5.21	4.85	5.21	7.43	4.12
GERMANTOWN	138,000	9.69	12.61	3.34	3.9	38.08	5.23	5.23	4.89	5.23	7.58	4.22
HINTZ	138,000	-	-	-	-	-	15.9	15.9	44.99	15.9	7.31	-
SPRING VALLEY WE	138,000	85.52	14.48	-	-	-	-	-	-	-	-	-
MAPLE	138,000	8.18	10.63	3.42	4.18	40.56	5.3	5.3	4.97	5.3	7.91	4.25
BUTLER	138,000	10.58	14.22	-	3.97	38.3	5.37	5.37	5.04	5.37	7.72	4.06

SUBSTATION	Cost Estimate (\$)	Cost Allocation (\$)										
		J1895	J1934	J1943	J1993	J2055	J2099	J2107	J2148	J2185	J2211	J2304
TAMARACK	488,000	52,710	69,992	15,698	18,434	174,836	25,419	25,419	23,691	25,419	36,292	20,090
GERMANTOWN	138,000	13,366	17,406	4,604	5,376	52,545	7,218	7,218	6,743	7,218	10,484	5,822
HINTZ	138,000	-	-	-	-	-	21,948	21,948	62,092	21,948	10,064	-
SPRING VALLEY_WE	138,000	118,016	19,984	-	-	-	-	-	-	-	-	-
MAPLE	138,000	11,293	14,676	4,726	5,771	55,966	7,313	7,313	6,865	7,313	10,894	5,870
BUTLER	138,000	14,600	19,626	-	5,478	52,852	7,407	7,407	6,955	7,407	10,671	5,597
Total WEC Affected System Upgrade Cost (\$) Allocated to Each		209,985	141,684	25,028	35,059	336,199	69,305	69,305	106,346	69,305	78,405	37,379

11.0 AVAILABLE APPENDIX DOCUMENTS (NOT ATTACHED)

Appendix A – Study Criteria, Methodology, and Assumptions

Appendix B – ATC Planning Criteria and Generation Facility Interconnection Guide

Appendix C – Interconnection Facility Project Diagrams_CEI

Appendix D – Network Upgrade Project Diagrams_CEI

Note: Project Diagrams were not developed for line upgrade projects.

Appendix E – Steady State Power Flow Results_CEI

Appendix G – Dynamic Stability Results_CEI

Appendix H - Affected System Study Results_CEI

Appendix I – MISO Deliverability Study Results_CEI

Appendix J – Assessed System Performance Reference

Appendix K – Voltage Stability Assessment_CEI

Appendix L - PSCAD Study Reports_CEI

Appendix M - Conditions to GIA

Appendix N - Short Circuit Data_CEI